# Default Risk Surprises and Banks' Loan Pricing

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# Motivation

**Credit markets exhibit substantial interest rate variation across firms and banks.**

**Most is explained by observables at loan issuance:** macroeconomic conditions, firm and loan characteristics and bank–firm relationships [Altavilla et al. (2024)]

**However, interest rate differentials persist among similar firms and loan products**

> **Cross-country evidence:** US [Amiti et al. (2026) WP], Italy [Bocola et al. (2026) WP], Portugal [Ferreira et al. (2024) WP], Brazil [Cavalcanti et al. (2023)] In France
>
> at least 15% of the variation in interest rates remains unexplained
>
> **Main drivers:** market imperfections (sources of imperfect competition and market power)

**However, these differences could also reflect information banks have and econometricians do not.**

# Frictions or information?

**The answer matters for:**

1. **Measurement:**
   - rate dispersion is informative of the forces at play in the market
2. **Policy:**
   - competition policy in local banking markets
   - value of public credit ratings and credit registries
   - firms' access to credit and consequences on financial health

# This paper: overview

**Empirics:** Using **French loan-level data**, I show that:

- Rate differentials $\tilde{r}_t$ are positively correlated with future increases in firms' credit risk $\tilde{c}_{t+k}$
- **Heterogeneous** across market segments: stronger for young, small firms, and in crises
- **Three mechanisms**: banks' information about borrower quality, endogenous sorting of risky firms into high rates, or a causal effect of paying a higher rate on financial health

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- **Three mechanisms**: banks' information about borrower quality, endogenous sorting of risky firms into high rates, ~~or a causal effect of paying a higher rate on financial health~~

**Theory:** A new model of loan pricing to disentangle the two mechanisms:

- search frictions + asymmetric information + imperfect common signal of firms' risk types
- Both information and search can deliver dispersion and the positive rate–risk relationship

**Calibration:**

- **Both information and search are necessary to account for the empirical patterns**
- Search frictions explain most of the rate variation
- In ex-ante safer markets, banks are informed and separate borrowers; in ex-ante riskier ones, they are uninformed and pool borrowers

# Literature

**Rate dispersion in loan markets**: Cavalcanti et al. (2021), Mazet-Sonilhac (2025), Amiti et al. (2026), Ferreira and Kozeniauskas (2026), Bocola et al. (2026)

→ I use **revisions of the firm's public rating** to measure banks' information (realized outcomes, more frequent than defaults)

**Models of price dispersion**: Varian (1980), Burdett and Judd (1983), Stahl (1989), Baye et al. (1992); **... of asymmetric information**: Akerlof (1970), Rothschild and Stiglitz (1976), Stiglitz and Weiss (1981); **... search and asymmetric information**: Guerrieri et al. (2010), Lester et al. (2019)

→ I combine **search frictions**, **asymmetric information** and an **imperfect signal** common to all banks (rather than screening through menus)

# Empirical Strategy

# Firm-level datasets

**1. Loan-level survey (NCE, 2006–2024)**: all new loans issued in the first month of each quarter, with the lender's id and firm-loan characteristics, all observable to the bank:

- **Loan**: rate, amount, maturity, loan type, fixed or floating rate, personal loan, loan over assets, pledged guarantees, several products or loans in the month
- **Firm**: age, young firm (0–3y), region, credit rating, group indicator, sector, individual entrepreneur; over assets: revenues, bank and non-bank debt, mobilized credit, short-term debt, undrawn credit lines

**2. Credit Registry**: outstanding stock of credit across bank–firm pairs

- new bank–firm relationships, department-level measures of competition

**3. Public Credit Ratings**: 3-year-ahead creditworthiness, assigned by Banque de France. Based on quantitative and qualitative data + repayment history + human expert evaluation.

- credit rating revisions after loan issuance, turned into default probabilities

⏩ Frequencies ⏩ Rating scale ⏩ Transitions ⏩ Why ratings

# Empirical strategy

1. **Loan rate differentials**: regress the rate of loan $l$ from lender $b$ to firm $i$ in month $t$ on observables and interacted fixed effects

$$r_{ilbt} = \alpha + \mathbf{x}'_{ilbt}\,\gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}$$

   $\mathbf{x}_{ilbt}$: loan and firm characteristics; $\mu_{bt\tau s}$: lender $\times$ month $\times$ loan type $\times$ 2-digit sector.

   The residual $\hat{\varepsilon}_{ilbt}$ is the unexplained rate component.

2. **Correlate it with future changes in credit risk**:
   - $\Delta P_{i,t+k} = P_{i,t+k} - P_{i,t+k-1}$: annual change in default probability between $k-1$ and $k$ years after issuance $\rightarrow$ $Corr(\hat{\varepsilon}_{ilbt}, \Delta P_{i,t+k})$

3. **In one step** (robust):
   - $r_{ilbt} = \alpha + \sum_{k=1}^{K} \beta_k \Delta P_{i,t+k} + \mathbf{x}'_{ilbt}\,\gamma + \mu_{bt\tau s} + \varepsilon'_{ilbt}$

Controls $\mathbf{x}_{ilbt}$ | Reg table | Dispersion and HHI

# Positive rate–risk correlation

$$r_{ilbt} = \alpha + \sum_{k=1}^{K} \beta_k \Delta P_{i,t+k} + \mathbf{x}'_{ilbt}\gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}$$

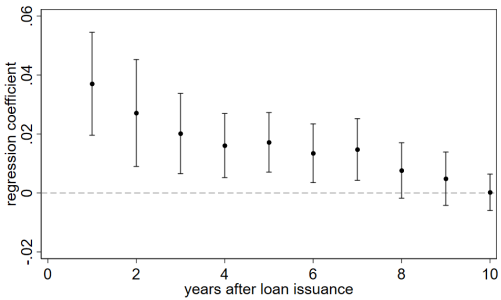



**Figure 3:** Rate residuals and changes in default probability, by year. All $K = 10$ revisions enter jointly; balanced sample; 95% intervals.

- Positive and significant, weaker at longer horizons

⏩ Magnitude ⏩ Additional Spec.

# Correlation: demographics

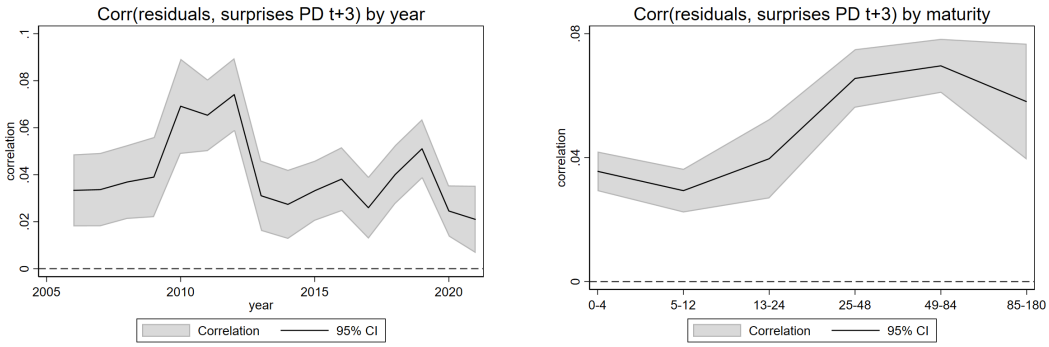



**Figure 4:** Correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability, both computed in the entire population and correlated by group $g$: by year (left) and by loan maturity (right). Bands: 95%.

# Correlation: demographics




**Figure 4:** (continued) By number of employees (left) and by firm age (right).

**↑ informational content when ↑ adverse selection.**

# Information, or a leverage effect?

Three possible mechanisms behind the correlation between rates and future credit outcomes:

1. **Information**: the bank knows the firm is riskier and charges it a higher rate
2. **Endogenous sorting**: risky firms sort into higher rates when prices are dispersed
3. **Leverage effect**: the high rate itself worsens the firm's fundamentals

**I rule out the third: fixed vs floating rates**

- Floating-rate borrowers are exposed to market rate increases [Ippolito et al. (2018)]
- When reference rates rise, are they more likely to be downgraded three years later?
- No, in the full sample and for loans of 2021q3–q4, when rates were at the zero lower bound and the 2022 increase was unexpected ⏩ Test ⏩ Full sample ⏩ 2021q3–q4

→ The reduced-form correlation cannot tell information from sorting: this calls for a model.

⏩ Rates and loan size

# Theory

# Environment

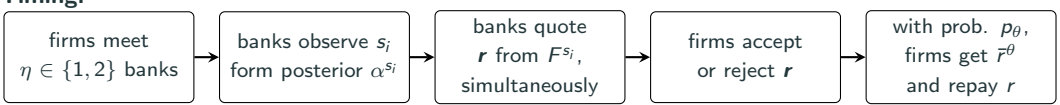
**Firms** with private information:

- adverse selection: risk types $\boldsymbol{\theta} \in \{\boldsymbol{g}, \boldsymbol{b}\}$ with shares $\mu_g$, $\mu_b$,
  Pr(success) $= p_\theta$ and reservation rate $\bar{r}^\theta$, where $p_b < p_g$ and $\bar{r}^g < \bar{r}^b$;
- search frictions: firms draw $\boldsymbol{\eta} \in \{\boldsymbol{1}, \boldsymbol{2}\}$ rate quotes, $\Pr(\eta = 2) = \pi$;
  accept the lowest quote if lower than $\bar{r}^\theta$, otherwise reject.

**Banks**:

- know neither $\theta$ nor $\eta$; choose $\boldsymbol{F(r)}$ to maximize profit $\Pi(r)$
- Common signal among banks: $\boldsymbol{s} \in \{s_g, s_b\}$, with shares $\Pr(s = s_g) = \sigma_g$ and precision $q$.
  - Banks form posteriors $\boldsymbol{\alpha^{s_i}} = \Pr(\theta = g \mid s = s_i)$, with $\alpha^{s_g} > \mu_g > \alpha^{s_b}$

**Timing:**



# Banks' profit maximization

**The bank chooses the CDF of rates $F^{s_i}(r)$ to maximize expected profits:**

$$\Pi^{s_i}(q,r) = \begin{cases} \underbrace{[1-\pi+\pi(1-F^{s_i}(r))]}_{\text{acceptance probability}}\underbrace{(r-c^{avg,s_i})}_{\text{profit per trade}} & \text{if } r \leq \bar{r}^g \\ \underbrace{(1-\alpha^{s_i})}_{\text{share bad}}\underbrace{[1-\pi+\pi(1-F^{s_i}(r))]}_{\text{acceptance probability}}\underbrace{(r-c_b)}_{\text{profit per trade}} & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

$\alpha^{s_i}$ is the probability of meeting good types given the signal $s_i$. with $\bar{r}^g < \bar{r}^b$

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$\alpha^{s_i}$ is the probability of meeting good types given the signal $s_i$. with $\bar{r}^g < \bar{r}^b$

- below $\bar{r}^g$, **both types accept**
- between $\bar{r}^g$ and $\bar{r}^b$, **only bad types accept**
- **Trade-off:** a higher rate raises the profit per trade and lowers the probability of acceptance

# Results Overview

1. **Equilibrium is dispersed**
   - search frictions introduce a trade-off between the rate and the probability of acceptance
   - when the signal is informative, banks offer on average higher rates to risky types

2. **Adverse selection determines where the bank lends**: depending on the signal and on the severity of adverse selection, banks either pool safe and risky borrowers at low rates below $\bar{r}^g$, or offer high rates only, which only risky types accept between $\bar{r}^g$ and $\bar{r}^b$

3. **The positive rate–risk correlation arises**:
   - when the signal is informative, banks offer higher rates to risky types
   - even without information, banks may post some rates in the pooling region and some in the separating region, leading to a partial separation between borrower types

⏩ Rate dispersion ⏩ Adverse selection ⏩ Examples

# Correlation between rates and riskiness

Information or search can each raise the correlation:

$$\rho(r, d) = \sqrt{n_g n_b}\, \frac{\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g]}{\sqrt{Var(r)}}$$

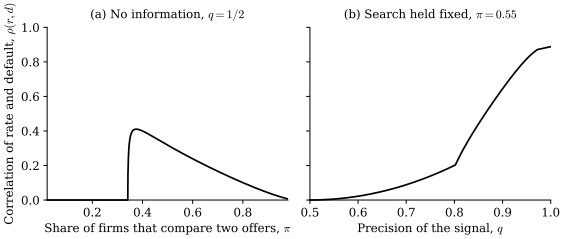



**Figure 8:** Panel (a) varies $\pi$ at fixed $q$; panel (b) varies $q$ at fixed $\pi$. Jumps: regime changes of the bad signal.

The correlation alone pins down neither $q$ nor $\pi$.

# Calibration

# Calibration: approach

Approach:

- **Model rate $\leftrightarrow$ rate residual**
- **Separately within each Banque de France rating class**

**Parameters taken directly from the data**, on firms whose rating later improves or worsens:

- shares $\mu_i$ and default probabilities $PD_i$ of safe and risky types, reservation rate $\bar{r}^b$ of risky types

**Three moments pin down three parameters**:

- moments: mean and variance of rate residuals, correlation between rates and subsequent risk
- parameters: share of firms with two offers $\pi$ (competition), signal precision $q$, reservation rate of good types $\bar{r}^g$ (adverse selection)

Approach Parameters Strategy

# Calibration: results

1. **Mild adverse selection within a credit rating**

   without information, banks **pool** firms and charge at most $\bar{r}^g$
   $\rightarrow$ the correlation in better classes requires informed banks

   - safest classes (3+ to 5+) $\rightarrow$ almost perfect signal $\hat{q} \geq 0.99$
   - riskiest classes (6 and 7) $\rightarrow$ uninformative signal $\hat{q} \approx 0.53$; banks pool

2. **Competition varies little across classes**: $\hat{\pi} \in [0.71, 0.77]$, declining slightly in riskier classes

3. **Information accounts for at most 5% of dispersion**

   - a lower bound: information is measured against realized outcomes, while banks price expected risk

**Both mechanisms are needed**; their relative importance depends on borrower risk.

⏩ Table 1 ⏩ Fit ⏩ Identification ⏩ Fixed $\pi$ ⏩ Defaults as bad types

# Conclusion

# Main takeaways

**Empirics:** rate differentials $\tilde{r}_t$ are positively correlated with future increases in credit risk $\tilde{c}_{t+k}$

- **Heterogeneous** across market segments: stronger for young, small firms, and in crises
- **Fixed vs floating rates**: rules out a causal effect of paying a higher rate on financial health

**Theory:** loan pricing with search frictions + asymmetric information + imperfect common signal of firms' risk types

- Information is worth more where asymmetric information is costly and search frictions are strong
- A more precise signal reduces (increases) the cost borne by safer (riskier) firms

**Calibration:** markets differ in **how precise the common signal** is: in ex-ante safer markets banks are informed and separate borrowers, in ex-ante riskier ones they pool them; search frictions are similar.

**Stability:** the calibrated signal precisions are an equilibrium of the acquisition game only if monitoring is about **nine times stronger** in the safest markets than in the riskiest.

⏩ Information acquisition

# Appendix

# France: interest rate dispersion

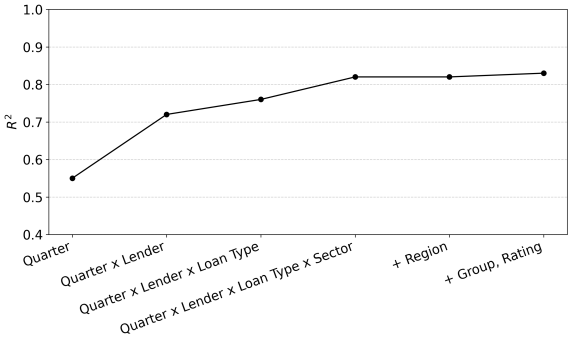



**Figure 1:** $R^2$ of the baseline regression, after gradually adding fixed-effects interactions and the other observables.

- Observables explain 83% of the variation: 17% remains unexplained
- Dispersion is large and time varying (approx. 1pp)

back | Reg Table | Approach | By bank | Monetary policy

# Baseline Specification

**Table B.2:** Data: Nouveaux Credits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Excludes credit lines and undefined maturity products. All credit and balance-sheet variables are expressed in terms of reported assets. Multiple products (dummy) equals one if the firm takes more than one product type in the quarter, counting the excluded products; Multiple loans, one product (dummy) equals one if it takes several loans in the quarter, all of the same type.

| | Total effective rate | | | | | |
|---|---|---|---|---|---|---|
| log.Loan quantity (K-Eur) | -0.2241*** | -0.1558*** | -0.1551*** | -0.1444*** | -0.1475*** | -0.0989*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Maturity (months) | -0.0006** | 0.0022*** | 0.0033*** | 0.0021*** | 0.0022*** | 0.0020*** |
| | (0.048) | (0.002) | (0.001) | (0.003) | (0.003) | (0.004) |
| Age | -0.0139*** | -0.0096*** | -0.0084*** | -0.0074*** | -0.0073*** | -0.0038*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Age squared | 0.0001*** | 0.0000*** | 0.0000*** | 0.0000*** | 0.0000*** | 0.0000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Young firm 0-3y (dummy) | -0.0538*** | -0.0414 | -0.0432 | -0.0198 | -0.0169 | -0.0677** |
| | (0.002) | (0.286) | (0.274) | (0.511) | (0.558) | (0.010) |
| New bank-firm relationship (dummy) | 0.1780*** | 0.0250 | 0.0190 | -0.0139 | -0.0148 | -0.0339** |
| | (0.000) | (0.526) | (0.633) | (0.309) | (0.273) | (0.024) |
| Floating rate (dummy) | 0.4331*** | 0.4889*** | 0.1677* | 0.1176 | 0.1160 | 0.1420 |
| | (0.000) | (0.000) | (0.052) | (0.201) | (0.201) | (0.125) |
| Personal loan (dummy) | -0.2967* | -0.0550 | 0.0397 | -0.0077 | -0.0053 | 0.0581 |
| | (0.051) | (0.701) | (0.626) | (0.932) | (0.951) | (0.466) |
| Individual entrepreneur (dummy) | 0.5594*** | 0.5740*** | 0.5365*** | 0.4364*** | 0.4237*** | 0.2432*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.001) |

back | explained dispersion | Firm FE | sample

# Baseline Specification, continued

| | | | | | | |
|---|---|---|---|---|---|---|
| Loan over assets | 0.1543** | 0.0985* | 0.0973* | 0.1050* | 0.1032* | 0.0474* |
| | (0.021) | (0.060) | (0.079) | (0.077) | (0.074) | (0.083) |
| Pledged guarantees | 0.7897*** | 0.1116 | -0.0221 | -0.2582** | -0.2378*** | -0.1769** |
| | (0.003) | (0.137) | (0.683) | (0.014) | (0.010) | (0.024) |
| Revenues | -0.0451*** | -0.0166 | -0.0119 | 0.0033 | 0.0049 | -0.0079 |
| | (0.000) | (0.176) | (0.350) | (0.599) | (0.453) | (0.136) |
| Bank Debt | 0.1596*** | 0.1232*** | 0.1202** | 0.1132*** | 0.1249*** | -0.0022 |
| | (0.000) | (0.008) | (0.012) | (0.001) | (0.001) | (0.912) |
| Non-Bank Debt | -0.0179 | -0.0601 | -0.0469 | 0.0034 | -0.0011 | -0.0200 |
| | (0.673) | (0.614) | (0.711) | (0.966) | (0.989) | (0.693) |
| Mobilised credit | -0.0106** | -0.0021 | -0.0012 | 0.0000 | -0.0002 | 0.0024 |
| | (0.030) | (0.472) | (0.724) | (0.992) | (0.944) | (0.307) |
| Short Term Debt | 0.0157 | -0.0018 | 0.0004 | 0.0202 | 0.0201 | -0.0126 |
| | (0.541) | (0.897) | (0.978) | (0.154) | (0.153) | (0.276) |
| Undrawn credit lines | -0.0139* | -0.0057 | -0.0043 | -0.0071 | -0.0071 | -0.0024 |
| | (0.063) | (0.227) | (0.337) | (0.119) | (0.110) | (0.447) |
| Multiple products (dummy) | -0.0256 | -0.0091 | -0.0368 | -0.0327* | -0.0301 | -0.0027 |
| | (0.201) | (0.731) | (0.130) | (0.092) | (0.100) | (0.815) |
| Multiple loans, one product (dummy) | -0.2182*** | -0.0656 | -0.0836 | -0.0427 | -0.0421 | -0.0073 |
| | (0.002) | (0.257) | (0.143) | (0.216) | (0.215) | (0.786) |
| Quarter FE | ✓ | | | | | |
| Lender × Quarter FE | | ✓ | | | | |
| Lender × Quarter × Loan Type FE | | | ✓ | | | |
| Lender × Quarter × Loan Type × 2digit Sector FE | | | | ✓ | ✓ | ✓ |
| Region FE | | | | | ✓ | ✓ |
| Group indicator, Rating FE | | | | | | ✓ |
| Observations | 457,868 | 457,868 | 457,868 | 457,868 | 457,868 | 457,868 |
| $R^2$ | 0.55 | 0.72 | 0.76 | 0.82 | 0.82 | 0.83 |

*p*-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

# The regression sample

| Rating Scale | N obs. | rate (mean) | rate (s.d.) | Quantity (mean, K-Eur) | Maturity (mean, months) | Var. Rate (in %) | Age (mean, years) | Loan/Assets (mean, in %) | Bank Debt/Assets (mean, in %) |
|---|---|---|---|---|---|---|---|---|---|
| 3++ | 7948 | 2.1 | 1.8 | 3754.5 | 59.0 | 35 | 39.0 | 5.4 | 19.6 |
| 3+ | 22681 | 2.2 | 1.8 | 2919.3 | 53.5 | 37 | 33.7 | 5.0 | 20.6 |
| 3 | 43073 | 2.3 | 1.8 | 2271.5 | 42.4 | 37 | 31.7 | 5.3 | 22.0 |
| 4+ | 71629 | 2.8 | 2.1 | 1092.2 | 34.8 | 39 | 28.1 | 6.2 | 23.0 |
| 4 | 112974 | 2.8 | 2.0 | 670.2 | 27.9 | 43 | 26.1 | 7.1 | 25.0 |
| 5+ | 106418 | 2.9 | 2.0 | 545.5 | 23.9 | 48 | 23.4 | 7.7 | 30.3 |
| 5 | 46510 | 3.5 | 2.2 | 781.8 | 17.9 | 51 | 23.1 | 8.3 | 31.0 |
| 6 | 13625 | 3.7 | 2.2 | 923.4 | 16.6 | 52 | 21.4 | 10.4 | 34.2 |
| 7 | 2657 | 4.2 | 2.3 | 134.7 | 14.8 | 62 | 19.7 | 8.6 | 32.7 |
| 8 | 1592 | 5.0 | 2.6 | 107.8 | 13.2 | 61 | 18.0 | 8.4 | 31.2 |
| 9 | 268 | 5.5 | 2.7 | 134.0 | 11.9 | 48 | 16.5 | 10.0 | 31.7 |
| P (liquidation) | 3716 | 5.0 | 2.0 | 114.5 | 3.4 | 26 | 21.8 | 5.7 | 33.4 |
| 0 (no rating) | 24777 | 3.2 | 2.1 | 1895.8 | 45.0 | 40 | 16.1 | 12.4 | 27.2 |
| **Total** | 457868 | 2.9 | 2.1 | 1098.1 | 30.5 | 44 | 25.8 | 7.3 | 26.5 |

**Table B.3:** Descriptive table of the loan-level regression sample. Source: Banque de France (Nouveaux Credits, COTA, SCR, FIBEN). The loan-level data refer to the first month of each quarter, so they are effectively monthly. Rate: total effective rate (including all fees).

back data

# Measuring dispersion: empirical approach

Extract the **unexplained component of rates** by controlling for:

- **Loan (l) characteristics** ($\beta$): log amount, maturity, floating-rate dummy, personal-loan dummy, loan over assets, pledged guarantees, multiple products and multiple loans in the quarter (dummies)
- **Firm (f) characteristics** ($\gamma$): age, age squared, young firm (0–3y), new bank–firm relationship, individual entrepreneur; balance sheet over assets: revenues, bank debt, non-bank debt, mobilised credit, short-term debt, undrawn credit lines
- **Interacted fixed effects** $\delta_{b,p,s,t}$: lender (b) $\times$ quarter (t) $\times$ loan type (p) $\times$ 2-digit sector (s)
- **Further fixed effects** $\lambda_f$: region, rating class, group indicator

$$\text{Rate}_{f,b,l,t} = \alpha + \beta \cdot \text{LoanChar}_{f,b,l,t} + \gamma \cdot \text{FirmChar}_{f,t} + \delta_{b,p,s,t} + \lambda_f + \varepsilon_{f,b,l,t}$$

table back

# Rate residuals distribution: variance and skewness

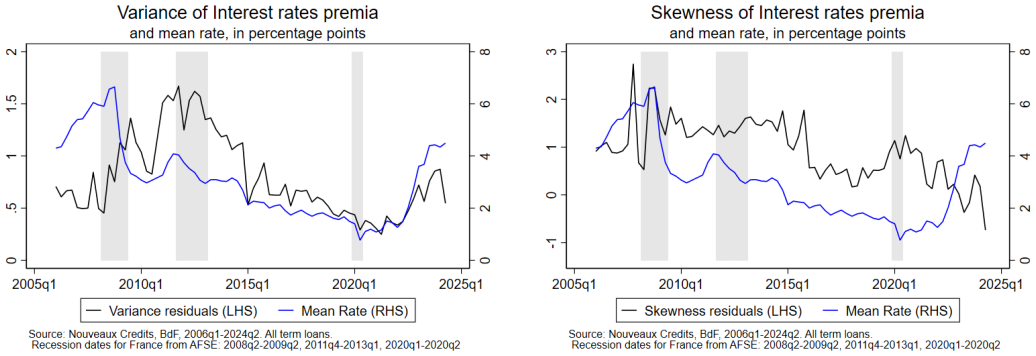
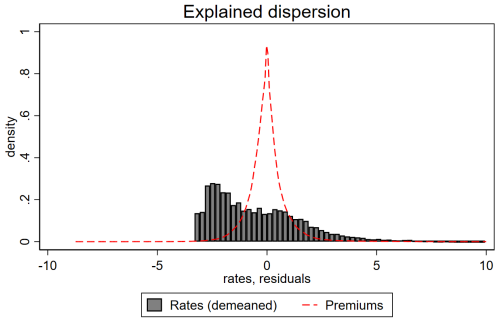


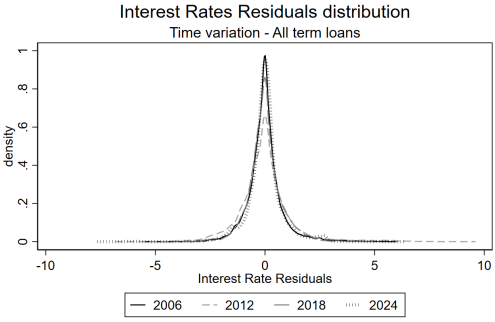

**Figure B.2:** Variance (left) and skewness (right) of rate residuals by quarter, black, left axis; mean rate, blue, right axis, in p.p. Shaded: French recessions (AFSE), 2008q2–2009q2, 2011q4–2013q1, 2020q1–2020q2. Data: Nouveaux Crédits, 2006q1–2024q2, all term loans.

Back

# Explained dispersion and time variation




Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.




Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.

**Figure B.3:** Left: rate residuals against actual rates, all sample. Right: distribution of residuals in selected years.

# Specification: firm fixed effects

| | Total effective rate | | | | | |
|---|---|---|---|---|---|---|
| log.Loan quantity (K-Eur) | -0.2337*** | -0.0215*** | -0.0200*** | -0.0183*** | -0.0231*** | -0.0182*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Maturity (months) | -0.0004*** | 0.0013*** | 0.0015*** | 0.0015*** | 0.0015*** | 0.0015*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Floating rate (dummy) | 0.4132*** | 0.1454*** | 0.2260*** | 0.2449*** | 0.2192*** | 0.2063*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Loan over assets | 0.0809*** | -0.0049 | -0.0053* | -0.0052* | -0.0047* | -0.0057** |
| | (0.000) | (0.105) | (0.063) | (0.060) | (0.086) | (0.035) |
| Quarter FE | ✓ | | | | | |
| Firm × Quarter FE | | ✓ | ✓ | ✓ | ✓ | |
| Lender FE | | | ✓ | | | |
| Lender × Quarter FE | | | | ✓ | ✓ | |
| Firm × Lender × Quarter FE | | | | | | ✓ |
| Loan Type FE | | | | | ✓ | ✓ |
| Observations | 590,621 | 197,343 | 197,328 | 196,048 | 196,042 | 156,211 |
| $R^2$ | 0.55 | 0.92 | 0.93 | 0.94 | 0.94 | 0.94 |

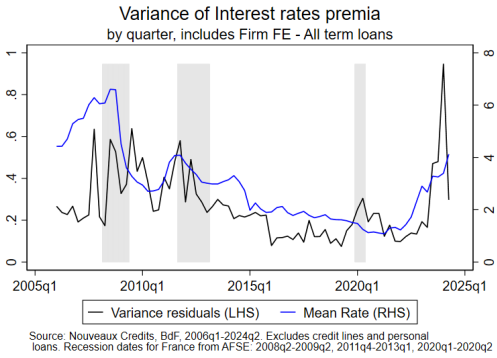
*p*-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

**Table B.4:** Data: Nouveaux Credits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. All credit and balance-sheet variables are expressed in terms of reported assets.

⏩ Back to Baseline ⏩ Variance and skewness

# Variance and skewness of rate premia - Firm FE



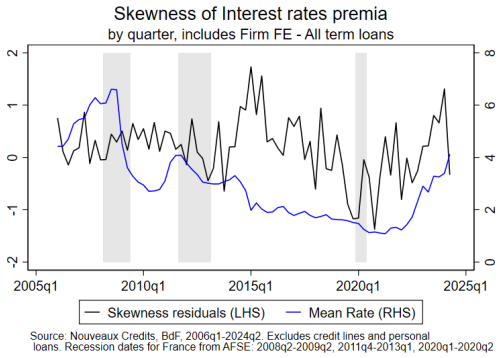




**Figure B.4:** Variance and skewness of residuals from the firm FE specification.

Back

# Within or Between bank dispersion?

Is the dispersion equal across banks, or are some banks "contributing more" to observed dispersion?

- I re-run the analysis separately for all banks and collect $R^2$

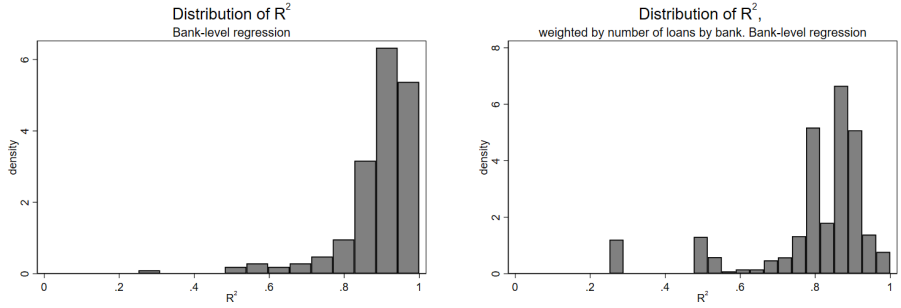



**Figure B.5:** Distribution of the $R^2$ of the regression, bank by bank, of the rate on loan characteristics, firm characteristics and time effects. Right panel: weighted by number of loans.

Back

# $R^2$ bank by bank: rating only

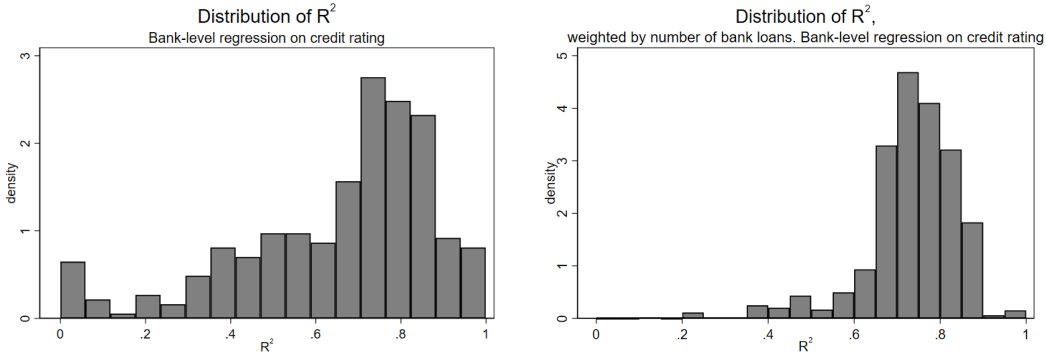



**Figure B.6:** Distribution of the $R^2$ of the regression, bank by bank, of the rate on the rating, with time and loan-type fixed effects. Right panel: weighted by number of loans.

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# Which banks use the BdF credit rating?

I compute the $\delta R^2$ explained by the credit rating.

- I re-run the analysis separately for all banks, controlling only for time and loan type FEs, and collect the $\delta R^2$

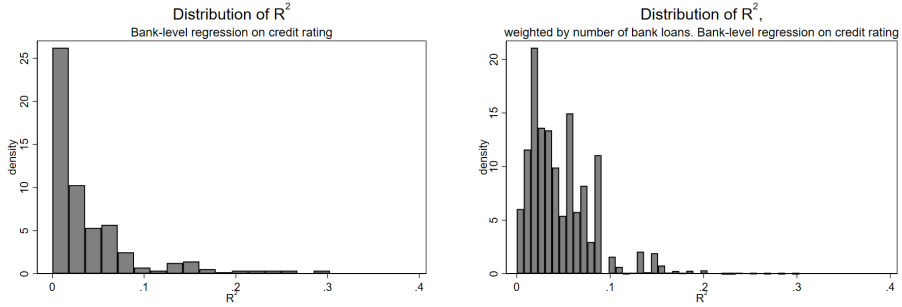



**Figure B.7:** Distribution of the increase in $R^2$ from adding the rating, bank by bank. Right panel: weighted by number of loans.

# Dispersion falls with market concentration

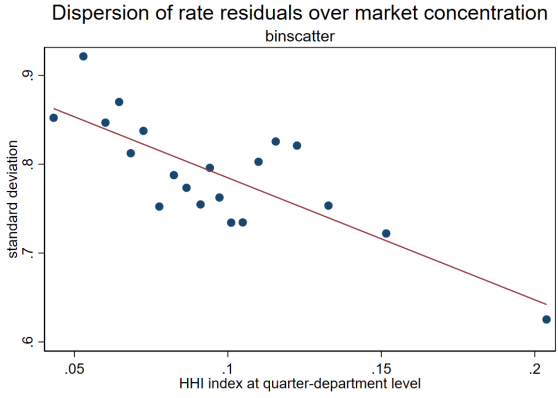



**Figure B.8:** Unexplained rate residuals' dispersion over HHI by department-quarter.

# Monetary Policy shock

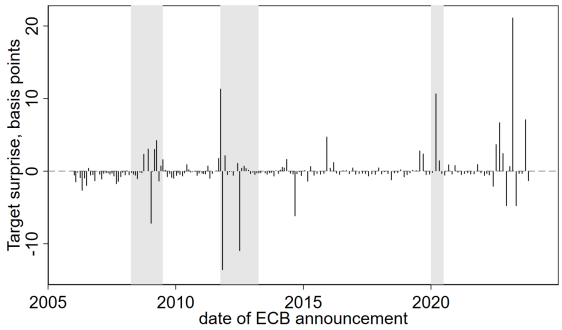



**Figure B.9:** Target surprise of Altavilla et al. (2019) at each ECB announcement, January 2006–October 2023, in basis points, scaled to a unit effect on the one-month OIS rate.

Specification

Back

# Effect of an increase of the policy rate

I compute and use high-frequency monetary policy shocks to test the effect of an increase in the funding cost of banks

In levels:[1]

$$Y_{lender,t} = \beta_0 + \beta_1 \cdot \text{MPshock}_t + \beta_2 Y_{lender,t-1} + \text{FE}_{lender} + \varepsilon$$

And in first differences:[2]

$$\Delta Y_{lender,t} = \gamma_0 + \gamma_1 \cdot \text{MPshock}_t + \text{FE}_{lender} + \epsilon$$

shock series results

[1] $\beta_1$ measures the effect of MP on the level of dispersion of premia, controlling for its lag. Assumes that there is inertia in the dispersion of rates over time.

[2] $\gamma_1$ measures the effect of MP on the change of dispersion of premia. No need to worry about trend.

# Effect of an increase of the policy rate

| | Dispersion | | Skewness | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| L1.rate shock | -0.0096*** | -0.0085*** | -0.0135*** | -0.0099* |
| | (0.000) | (0.003) | (0.010) | (0.093) |
| L2.rate shock | | 0.0001 | | -0.0227*** |
| | | (0.979) | | (0.000) |
| L3.rate shock | | -0.0017 | | -0.0208*** |
| | | (0.568) | | (0.000) |
| L1.SD / L1.Skew | 0.2813*** | 0.2799*** | 0.1294*** | 0.1171*** |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| Lender FE | ✓ | ✓ | ✓ | ✓ |
| Observations | 7,189 | 7,077 | 7,189 | 7,077 |
| $R^2$ | 0.40 | 0.41 | 0.23 | 0.24 |

*p*-values in parentheses

* p<0.10, ** p<0.05, *** p<0.01

**Table B.5:** Response of the distribution of rate residuals to monetary policy. Each observation is a lender-quarter. Dispersion and Skewness are the standard deviation and the skewness of the lender's rate residuals from the baseline regression in the quarter. Each column estimates the specification on the previous slide, with one lag of the shock in columns (1) and (3) and three lags in columns (2) and (4). The shock is the Target factor of Altavilla et al. (2019) summed over the ECB announcements in the quarter, in standard deviations of the announcement-level factor (one standard deviation is 2.7 basis points of the one-month OIS rate). Data: Nouveaux Crédits (BdF), 2006q1–2024q2, quarterly shocks. The loan-level data refer to the first month of each quarter, so they are effectively monthly.

Back

# Measuring the firm's type: rating transitions

**3. Banque de France public ratings (COTA):**

Forward-looking over 3 years; quantitative and qualitative data, history and payment incidents. I turn ratings into default probabilities with historical default frequencies PD Rating scale Transitions

**I proxy the firm's type at issuance with how its default probability evolves afterwards.**

1. three-year default frequency in each rating class → default probability per class frequencies
2. firm upgraded or downgraded → $\Delta P_{i,t+k}$ = PD of the new class − PD of the old one classes

**Why public ratings:**

- **Direct measure of creditworthiness**: balance sheets, firm history, interviews — not an ex-post proxy (returns, growth)
- **Variation**: annual reviews give many transitions per firm; actual defaults are rare
- **Realized outcome**, not a bank's expectation: internal ratings explain rates [Amiti et al., 2026] but go untested against default, and may reflect capital requirements

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# Default frequencies by rating

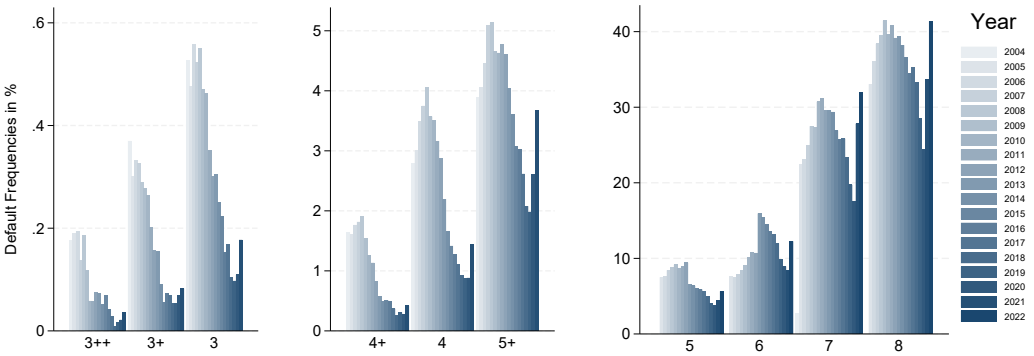



**Figure 2:** 3-year ahead default frequencies, by rating class and year (2004-2022). All firms with a rating at 1 January of the year; default by 31 December two years later, as in the Banque de France's published reports.

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# BdF Credit Rating

Three possible entries.

1. **Credit Rating** (= Probability of default)

   Each class maps to its three-year default frequency, from the 2021 cohort ⏩ frequencies

2. **Actual Default**

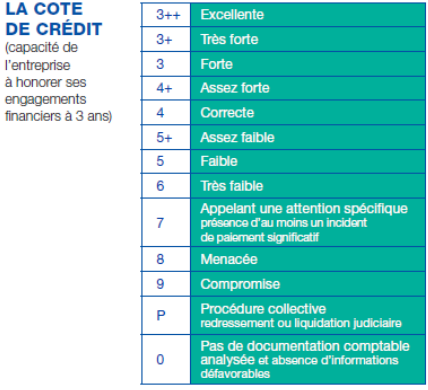
   Takes value "P"

3. **Not enough information**

   Takes value "0"

⏩ scale ⏩ transitions ⏩ back

# Banque de France rating scale



**LA COTE DE CRÉDIT**
(capacité de l'entreprise à honorer ses engagements financiers à 3 ans)

| | |
|---|---|
| 3++ | Excellente |
| 3+ | Très forte |
| 3 | Forte |
| 4+ | Assez forte |
| 4 | Correcte |
| 5+ | Assez faible |
| 5 | Faible |
| 6 | Très faible |
| 7 | Appelant une attention spécifique<br>présence d'au moins un incident de paiement significatif |
| 8 | Menacée |
| 9 | Compromise |
| P | Procédure collective<br>redressement ou liquidation judiciaire |
| 0 | Pas de documentation comptable analysée et absence d'informations défavorables |

**Figure A.1:** Source: Banque de France. Credit rating scale until 2022.

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# Rating transitions

**Table A.1:** Three-year rating transitions, 2021 cohort

| | 31 Dec. 2023 (% of row) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 Jan. 2021 | 3++ | 3+ | 3 | 4+ | 4 | 5+ | 5 | 6 | 7 | 8 | P | 0 | Exit | Firms |
| 3++ | **46.4** | 15.3 | 15.1 | 9.4 | 2.6 | 1.8 | 0.3 | 0.7 | 0.0 | 0 | 0.0 | 8.4 | 0 | 9,365 |
| 3+ | 14.8 | **20.4** | 25.7 | 22.6 | 5.8 | 2.2 | 0.4 | 0.6 | 0.0 | 0 | 0.1 | 7.4 | 0 | 16,013 |
| 3 | 5.5 | 9.9 | **26.6** | 32.6 | 11.6 | 3.9 | 0.8 | 0.7 | 0.0 | 0.0 | 0.1 | 8.1 | 0 | 26,543 |
| 4+ | 2.6 | 3.4 | 14.8 | **35.3** | 22.4 | 8.1 | 1.4 | 0.9 | 0.1 | 0.0 | 0.3 | 10.7 | 0 | 35,727 |
| 4 | 0.9 | 1.2 | 6.5 | 26.3 | **33.0** | 14.5 | 3.5 | 1.8 | 0.3 | 0.2 | 0.9 | 10.9 | 0.0 | 53,173 |
| 5+ | 0.3 | 0.5 | 1.5 | 8.5 | 15.8 | **22.1** | 7.9 | 5.0 | 0.6 | 0.3 | 2.6 | 34.8 | 0.0 | 88,043 |
| 5 | 0.0 | 0.1 | 0.2 | 1.0 | 2.7 | 4.5 | **61.1** | 2.8 | 0.4 | 0.3 | 4.3 | 21.6 | 1.0 | 126,013 |
| 6 | 0.1 | 0.1 | 0.2 | 0.7 | 2.0 | 4.5 | 46.0 | **6.9** | 0.7 | 0.4 | 8.4 | 29.5 | 0.4 | 54,905 |
| 7 | 0 | 0 | 0.1 | 0.3 | 0.8 | 1.8 | 5.4 | 5.3 | **7.6** | 6.7 | 27.7 | 44.0 | 0.3 | 4,639 |
| 8 | 0 | 0 | 0 | 0.2 | 0.8 | 1.2 | 3.7 | 5.1 | 6.9 | **11.1** | 33.6 | 37.1 | 0.3 | 2,923 |
| 9 | 0 | 0 | 0 | 0 | 0.5 | 0.9 | 2.7 | 6.3 | 4.1 | 15.8 | 39.2 | 30.2 | 0.5 | 222 |

Notes: Firms rated by the Banque de France on 1 January 2021, by credit rating (rows), and their rating on 31 December 2023 (columns). Each cell is the percentage of the row's firms that hold the column's rating at the horizon, so each row sums to 100; the diagonal, in bold, is the share still in the same class. P: the firm is in insolvency proceedings. 0: the firm is in the file but has no rating based on financial statements. Exit: the firm no longer appears in the rating file. Ratings issued after the 2022 scale reform are converted back to the pre-reform scale; the conversion merges the new class 9 into 8, so the horizon has no class 9. Firms: number of firms in the row's class on 1 January 2021.

◂ Back

# Magnitude: a quarter of the break-even spread

**Estimate**: a one-point rise in the three-year default probability one year after issuance goes with an unexplained 3.7 bp rate premium at issuance ($\beta_1$, s.e. 0.9).

**Benchmark**: the bank loses $\mathrm{LGD}$ when the firm defaults; constant one-year default probability $\lambda$, so $P_T = 1 - (1-\lambda)^T$. The bank breaks even at the annual spread

$$s \;=\; \mathrm{LGD}\,\frac{\lambda}{1-\lambda} \qquad \Longrightarrow \qquad \frac{\partial s}{\partial P_T} \;\approx\; \frac{\mathrm{LGD}}{T},$$

whatever the discount factor, the maturity and the repayment schedule.

**Comparison**: median bank recovery of 56% in French defaults [Davydenko and Franks, 2008] gives $\mathrm{LGD} = 0.44$; with $T = 3$, break-even is 15 bp.

- $\beta_1$ is 25% of break-even
- Same in one-year units: 11 bp against 44 bp per point of one-year default probability
- The benchmark predicts $\beta_k$ falls with $k$: a revision in year $k$ raises the loss only on the balance still outstanding

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# Correlation residuals vs actual default

$$r_{ilbt} = \alpha + \sum_{j=1}^{k} \beta_j^D D_{i,t+j} + \mathbf{x}'_{ilbt}\gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \qquad D_{i,t+j} = 100 \text{ if default in year } j$$

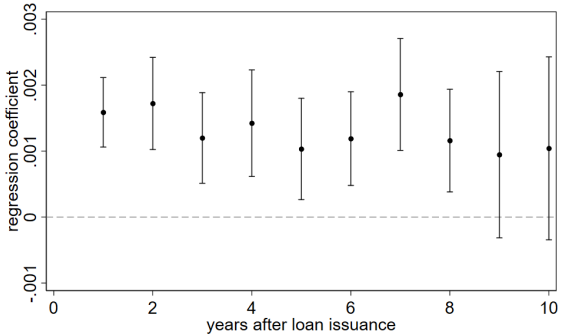



**Figure B.10:** Correlation between rate residuals and actual default, by horizon. One regression per $k$, reporting $\beta_k^D$; standard errors clustered by lender and quarter.



by rating | 1y prob default

# Actual default: by ex-ante rating

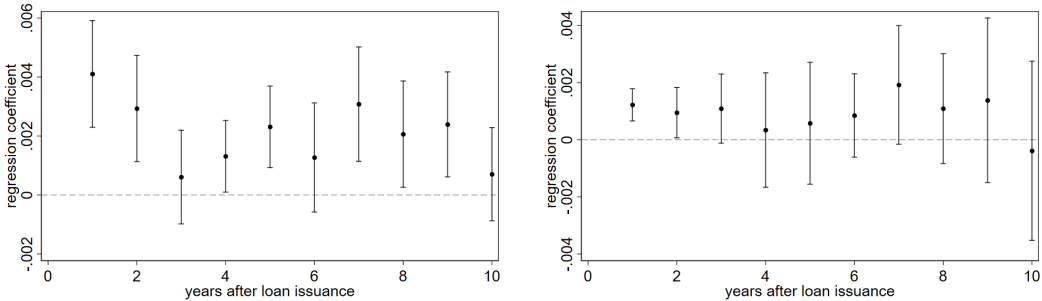



**Figure B.11:** Correlation between rate residuals and actual default. Left: good classes, 3++ to 4+ (ex-ante PD$<$ 0.62%). Right: bad classes, 5 to 9 (PD$>$ 3.45%). Classes 4 and 5+ are left out.

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# The rate predicts annual revisions: balanced sample

$$\Delta P_{i,t+k} = \alpha_k + \theta_k\, r_{ilbt} + \sum_{j<k} \delta_{kj} \Delta P_{i,t+j} + \mathbf{x}'_{ilbt}\gamma_k + \mu_{bt\tau s} + \varepsilon_{ilbt}$$

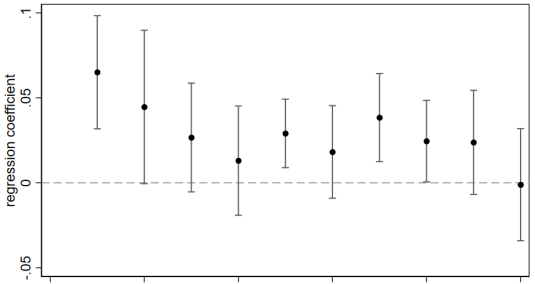
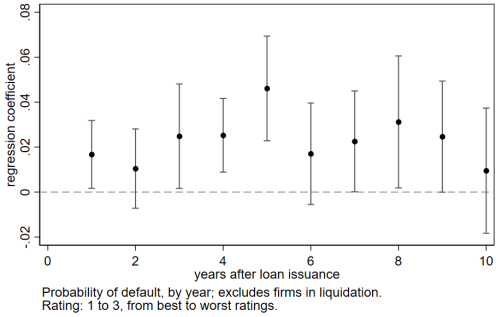



**Figure B.12:** Coefficient $\theta_k$ on the rate, year-on-year revisions, past revisions controlled, balanced sample. $\theta_1 = 0.064$ (s.e. 0.017). Bars: 95% intervals, clustered by lender.

# Annual revisions, balanced sample: good vs bad ratings



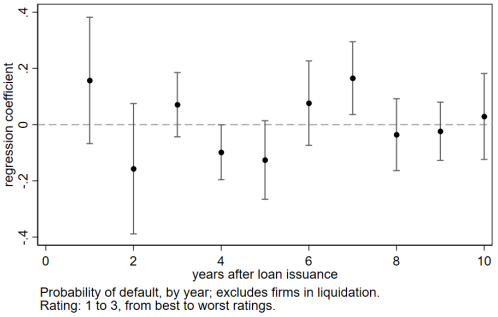




**Figure B.13:** Coefficient $\theta_k$ on the rate. Left: good classes. Right: bad classes. Note the different vertical scales.

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# The rate predicts annual revisions: unbalanced sample

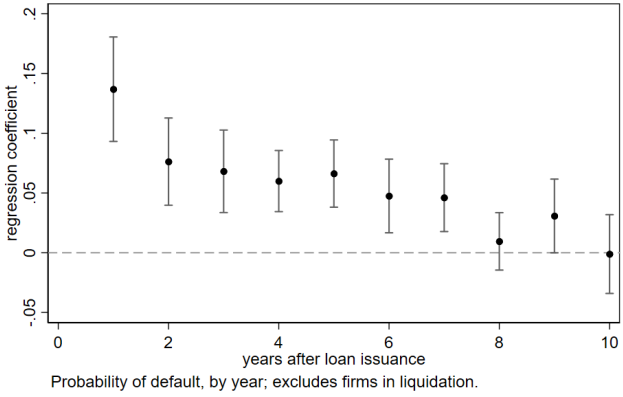
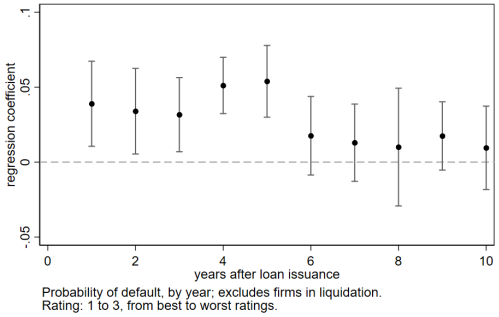



**Figure B.14:** Coefficient $\theta_k$ on the rate, same specification as Figure B.12, unbalanced sample. $\theta_1 = 0.14$ (s.e. 0.022), falling to zero by year 10.

# Annual revisions, unbalanced sample: good vs bad ratings



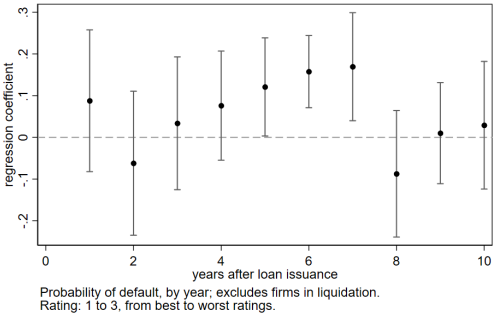




**Figure B.15:** Coefficient $\theta_k$ on the rate. Left: good classes. Right: bad classes. Note the different vertical scales.

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# The rate predicts cumulative revisions

$$\Delta P_{i,t\to t+k} = \alpha_k + \theta_k\, r_{ilbt} + \mathbf{x}'_{ilbt}\gamma_k + \mu_{bt\tau s} + \varepsilon_{ilbt}$$

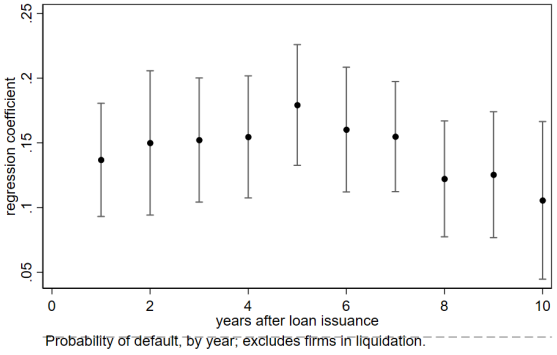
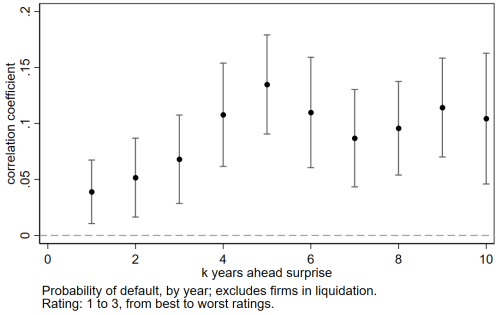



**Figure B.16:** Coefficient $\theta_k$ on the rate, cumulative change from issuance. $\theta_5 = 0.18$ (s.e. 0.023), and 0.11 at ten years.

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# Cumulative revisions: good vs bad ratings



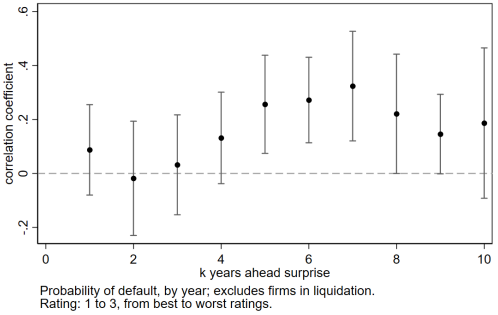




**Figure B.17:** Coefficient $\theta_k$ on the rate. Left: good classes, 0.039 at one year to 0.14 at five. Right: bad classes. Note the different vertical scales.

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# Correlation: stronger for better rated firms

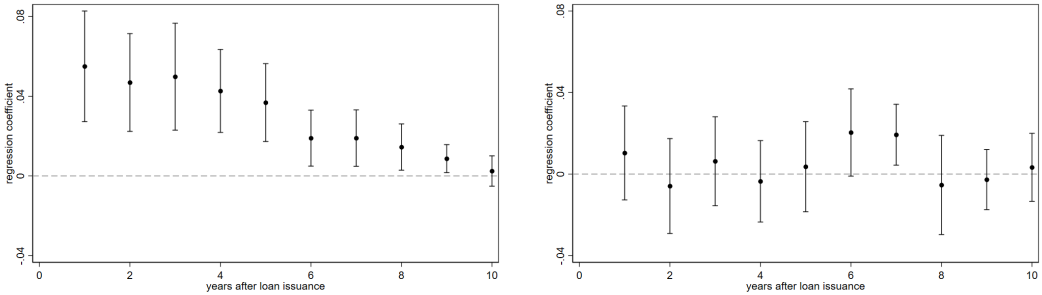



**Figure B.18:** Correlation between rate residuals and changes in default probability. Left: good classes, 3++ to 4+ (ex-ante PD$<$ 0.62%). Right: bad classes, 5 to 9 (PD$>$ 3.45%). Classes 4 and 5+ are left out.

**Predictive power of rate residuals varies by ex-ante riskiness.**

# Correlation: does not depend on loan size or bank debt

**Mechanical effect** → stronger if the loan is large! Focusing on the better rated firms (low PD):

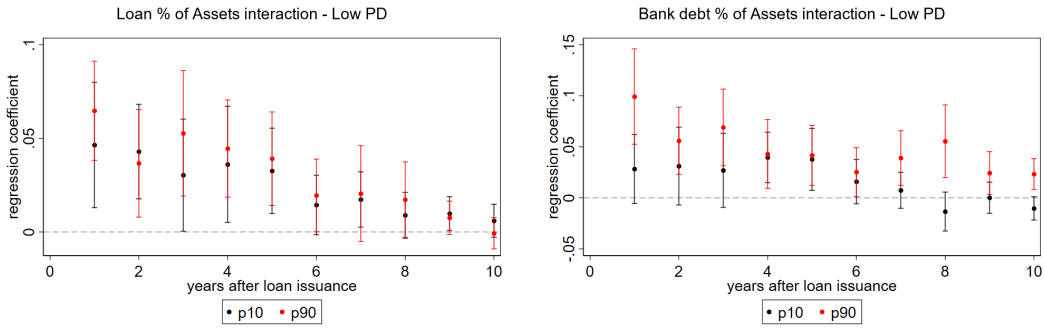



**Figure B.19:** Interactions of the change in default probability, 10th vs 90th percentile. Left: loan size over assets. Right: bank debt over assets. Classes 3++ to 4+; balanced sample, firms in liquidation excluded.

# Correlation: stronger for younger firms

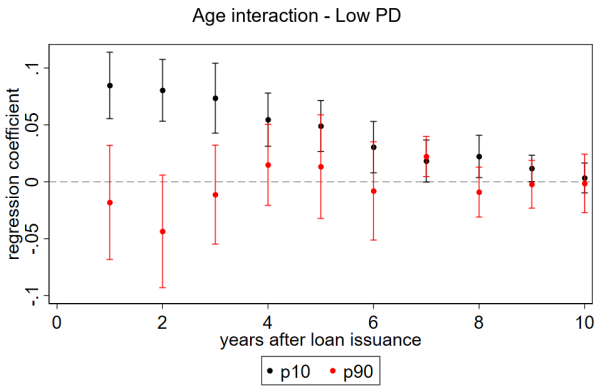



**Figure B.20:** Interaction of the change in default probability with firm age, below the 10th vs above the 90th percentile. Classes 3++ to 4+.

# Information, or a leverage effect?

A positive correlation admits two readings:

**Information** — the bank recognizes the riskier type and prices it

**Leverage** — the high rate mechanically increases the size of debt

Two approaches:

1. I explore how the **correlation varies in the cross-section**
2. **Fixed vs variable rates** test rules out $\tilde{r}_t \to \tilde{c}_{t+k}$

# Test: do higher rates induce rating worsening?

**Design.** Compare **variable**- and **fixed**-rate borrowers; when reference rates rise, are variable-rate borrowers more likely to experience a rating worsening?

**Assumptions.**

1. Exogeneity: the increase is set by aggregate conditions and monetary policy, so it is exogenous to the individual firm's fundamentals.
2. Sorting on rate expectations: (i) I show that the contract choice is predominantly driven by banks' preferences (ii) I repeat the exercise at the end of 2021 ⏩ Self-selection
3. Parallel trends: Absent the rate increase, rating transitions of fixed- and floating-rate borrowers would have evolved in the same way. ⏩ Validity

# Test: do higher rates induce rating worsening?

**Design.** Compare **variable**- and **fixed**-rate borrowers; when reference rates rise, are variable-rate borrowers more likely to experience a rating worsening?

$$\Delta PD_{i,1\to3} = \beta\,\Delta\mathrm{Ref}_{l,1\to2} + \delta\left(\Delta\mathrm{Ref}_{l,1\to2}\times \mathrm{Share}_{i,1}\right) + \mathbf{x}'_{ilbt}\,\gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}$$

$\Delta PD_{i,1\to3} = 100\times(P_{i,3}-P_{i,1})$; $\mathrm{Share}_{i,1}$ = loan- or bank-debt-over-assets.
$\Delta\mathrm{Ref}_{l,1\to2}$ is loan $l$'s **own** reference rate change: each floating loan is tied to its own index (EONIA, Euribor 1m/3m/1y, TMO), a fixed-rate loan to zero.
Maturity $\geq$ 30 months; SEs two-way clustered by lender and month.

**Variation** within a lender $\times$ month $\times$ loan-type $\times$ sector cell: fixed vs floating, and *which index* a floating loan carries.

Back

# Test: do higher rates induce rating worsening?

| | Default Probability | | | | | | |
|---|---|---|---|---|---|---|---|
| Floating rate (dummy) | 0.1176** | | 0.1407 | 0.1039** | | | |
| | (0.014) | | (0.127) | (0.026) | | | |
| cum dRate end y2 | | 0.0433 | | | 0.0374 | | |
| | | (0.229) | | | (0.254) | | |
| annual dRate end y1 | | | 0.0812 | | | 0.0524 | 0.0425 |
| | | | (0.600) | | | (0.711) | (0.773) |
| annual dRate end y2 | | | 0.0567 | | | 0.0363 | 0.0128 |
| | | | (0.330) | | | (0.551) | (0.879) |
| Floating rate (dummy) × Loan over assets | | | | 0.1932 | | | |
| | | | | (0.370) | | | |
| cum dRate end y2 × Loan over assets | | | | | 0.0711 | | |
| | | | | | (0.264) | | |
| annual dRate end y1 × Loan over assets | | | | | | -0.1330 | |
| | | | | | | (0.763) | |
| annual dRate end y2 × Loan over assets | | | | | | 0.0905 | |
| | | | | | | (0.148) | |
| annual dRate end y1 × Bank Debt share | | | | | | | 0.0058 |
| | | | | | | | (0.980) |
| annual dRate end y2 × Bank Debt share | | | | | | | 0.0986 |
| | | | | | | | (0.623) |
| Loan over assets | 0.2250** | 0.2002** | 0.2048** | 0.1808* | 0.1894* | 0.1849* | 0.1929* |
| | (0.030) | (0.035) | (0.035) | (0.094) | (0.054) | (0.059) | (0.054) |
| Bank Debt share | 0.0846** | 0.1139** | 0.1134** | 0.0850** | 0.1118** | 0.1114** | 0.1127** |
| | (0.040) | (0.019) | (0.019) | (0.034) | (0.022) | (0.023) | (0.025) |
| Lender × Month × Loan Type × 2digit Sector FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Region FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| All other observables | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 102,463 | 96,414 | 96,414 | 102,463 | 96,414 | 96,414 | 96,414 |
| $R^2$ | 0.42 | 0.40 | 0.40 | 0.42 | 0.40 | 0.40 | 0.40 |

*p-values in parentheses*

* p< 0.10, ** p< 0.05, *** p< 0.01

**Table B.6:** Data: Nouveaux Crédits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Term loans, at least 30-month maturity, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

# Rate exercise: 2021q3–q4, 2022 increase only

| | Default Probability - 2021q3-2021q4 only | | | | | | |
|---|---|---|---|---|---|---|---|
| Floating rate (dummy) | -0.5698 | | -6.1935 | -0.5121 | | | |
| | (0.169) | | (0.193) | (0.186) | | | |
| annual dRate end y2 | | | 1.8860 | | | -0.3828* | -0.4445 |
| | | | (0.261) | | | (0.095) | (0.218) |
| Floating rate (dummy) × Loan over assets | | | | -0.6271 | | | |
| | | | | (0.249) | | | |
| annual dRate end y2 × Loan over assets | | | | | | -0.3410 | |
| | | | | | | (0.228) | |
| annual dRate end y2 × Bank Debt share | | | | | | | 0.0555 |
| | | | | | | | (0.915) |
| Loan over assets | -0.5329 | -1.1142 | -1.1243 | -0.1584 | -0.6316 | -0.6318 | -1.1395 |
| | (0.374) | (0.168) | (0.216) | (0.740) | (0.224) | (0.223) | (0.177) |
| Bank Debt share | 0.0986 | -0.2614 | -0.2672 | 0.1401 | -0.2234 | -0.2234 | -0.2780 |
| | (0.725) | (0.316) | (0.448) | (0.631) | (0.375) | (0.375) | (0.364) |
| Lender × Month × Loan Type × 2digit Sector FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Region FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| All other observables | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 4,324 | 4,089 | 4,089 | 4,324 | 4,089 | 4,089 | 4,089 |
| $R^2$ | 0.41 | 0.41 | 0.41 | 0.41 | 0.41 | 0.41 | 0.41 |

*p*-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

**Table B.8:** Data: Nouveaux Crédits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

⏩ 2022–23 increase

# Test: do higher rates induce rating worsening?

**Result:** $\beta$ and $\delta$ are not significant.

- a rise in reference rates does not induce a rating worsening
- suggests that the correlation between rate residuals and rating changes **is not mechanically driven by leverage**

Two approaches:

✓ **Information** — the bank recognizes the riskier type and prices it

✗ **Leverage** — the high rate mechanically increases the size of debt

# Fixed vs floating: loans of 2021q3–q4, rate change through 2023

| | Default Probability - 2021q3-2021q4 only | | | | | | |
|---|---|---|---|---|---|---|---|
| Floating rate (dummy) | -0.5698 | | -19.6243 | -0.5121 | | | |
| | (0.169) | | (0.479) | (0.186) | | | |
| annual dRate end y2 | | | 5.0816 | | | 0.0935 | 0.6439 |
| | | | (0.477) | | | (0.943) | (0.316) |
| annual dRate end y3 | | | 2.6790 | | | -0.7103 | -1.5704 |
| | | | (0.563) | | | (0.718) | (0.179) |
| Floating rate (dummy) × Loan over assets | | | | -0.6271 | | | |
| | | | | (0.249) | | | |
| annual dRate end y2 × Loan over assets | | | | | | 5.0753 | |
| | | | | | | (0.781) | |
| annual dRate end y2 × Bank Debt share | | | | | | | -0.6506 |
| | | | | | | | (0.424) |
| annual dRate end y3 × Loan over assets | | | | | | -7.6615 | |
| | | | | | | (0.769) | |
| annual dRate end y3 × Bank Debt share | | | | | | | 0.9643 |
| | | | | | | | (0.155) |
| Loan over assets | -0.5329 | -1.1153 | -1.1298 | -0.1584 | -0.6351 | -0.6550 | -1.1379 |
| | (0.374) | (0.168) | (0.233) | (0.740) | (0.216) | (0.287) | (0.283) |
| Bank Debt share | 0.0986 | -0.2617 | -0.2685 | 0.1401 | -0.2242 | -0.2431 | -0.2795 |
| | (0.725) | (0.313) | (0.569) | (0.631) | (0.369) | (0.431) | (0.674) |
| Lender × Month × Loan Type × 2digit Sector FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Region FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| All other observables | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 4,324 | 4,089 | 4,089 | 4,324 | 4,089 | 4,089 | 4,089 |
| $R^2$ | 0.41 | 0.41 | 0.41 | 0.41 | 0.41 | 0.41 | 0.41 |

*p*-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

**Table B.9:** Data: Nouveaux Crédits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Back

# Validity checks

- **Mitigate self-selection issue:**
  - contract choice is predominantly driven by supply-side factors
  - end of ZLB period, firms do not expect a rate hike
- **Balance table** (in 2021):
  - some characteristics differ (variable: fewer firms, larger size, older, riskier, larger financial debt)
  - All of those are included as controls
- **Parallel trends test**:
  - visually: rating transitions for firms taking $> 2.5y$ loans do not differ significantly ⏩ Figure
  - Pre-treatment betas are close to zero (not ***)

⏩ back

# Parallel trends: fixed vs. floating

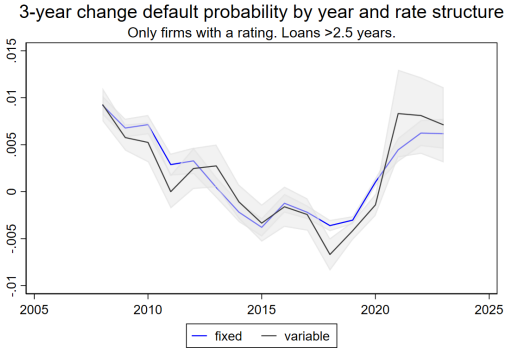



**Figure B.21:** Mean three-year change in the default probability by year, for fixed- and floating-rate borrowers separately. Term loans over 2.5 years, firms with a rating. Shaded: 95% confidence bands.

The two groups track each other through the pre-period. Back

# Can variable/fixed rate contract be predicted?

Focusing on firms with a rating at the time of the loan, I estimate:

$$Floating_{loan} = \beta X^{observables} + \delta_{t,b,p,s} + \varepsilon_{loan}$$

where $Floating_{loan}$ is a dummy for floating-rate loans and $\delta_{t,b,p,s}$ is the fixed effects interaction term for month, bank, loan purpose and firm sector.

**Table B.7:** Variance Decomposition of Floating Rate Choice. Data: Nouveaux Crédits (BdF). The loan-level data refer to the first month of each quarter, so they are effectively monthly.

| **Fixed Effects Included** | **Adjusted $R^2$** |
|---|---|
| Quarter | 0.04 |
| + Lender | 0.68 |
| + Loan Type | 0.89 |
| + Firm Sector | 0.92 |

The contract choice is predominantly driven by supply-side and sectoral factors rather than idiosyncratic firm risk back

# Rates do not rise with loan size

**Table B.10:** Loan rates and loan size. Each column reruns the baseline regression (Table B.2, column 6) keeping one measure of loan size: loan over assets in column (1), log loan quantity in column (2). Data: Nouveaux Crédits (BdF), 2006q1–2024q2, excluding credit lines and undefined maturity products. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Variables defined as in Table B.2.

| | Total effective rate | |
|---|---|---|
| | (1) | (2) |
| log.Loan quantity (K-Eur) | | -0.0960*** |
| | | (0.000) |
| Loan over assets | -0.0437 | |
| | (0.735) | |
| Maturity (months) | 0.0012** | 0.0020*** |
| | (0.034) | (0.005) |
| Age | -0.0044*** | -0.0038*** |
| | (0.000) | (0.000) |
| Age squared | 0.0000*** | 0.0000*** |
| | (0.000) | (0.000) |
| Young firm 0-3y (dummy) | -0.0799*** | -0.0653** |
| | (0.006) | (0.015) |
| New bank-firm relationship (dummy) | -0.0422*** | -0.0379** |
| | (0.009) | (0.010) |
| Floating rate (dummy) | 0.0907 | 0.1479 |
| | (0.335) | (0.107) |
| Personal loan (dummy) | 0.1023 | 0.0741 |
| | (0.136) | (0.309) |

Back

# Rates do not rise with loan size, continued

| | | |
|---|---|---|
| Individual entrepreneur (dummy) | 0.2617*** | 0.2434*** |
| | (0.001) | (0.001) |
| Pledged guarantees | -0.4114*** | -0.3477*** |
| | (0.001) | (0.000) |
| Revenues | 0.0054 | -0.0030 |
| | (0.464) | (0.630) |
| Bank Debt | -0.0270 | -0.0249 |
| | (0.400) | (0.411) |
| Non-Bank Debt | -0.0301 | -0.0296 |
| | (0.689) | (0.701) |
| Mobilised credit | 0.0413 | 0.0714** |
| | (0.158) | (0.013) |
| Short Term Debt | -0.2244*** | -0.1596*** |
| | (0.000) | (0.002) |
| Undrawn credit lines | -0.1253 | -0.0255 |
| | (0.146) | (0.833) |
| Multiple products (dummy) | 0.0016 | 0.0024 |
| | (0.887) | (0.843) |
| Multiple loans, one product (dummy) | 0.0124 | -0.0074 |
| | (0.608) | (0.781) |
| Lender × Quarter × Loan Type × 2digit Sector FE | ✓ | ✓ |
| Region FE | ✓ | ✓ |
| Group indicator, Rating FE | ✓ | ✓ |
| Observations | 449,973 | 451,305 |
| $R^2$ | 0.82 | 0.83 |

*p*-values in parentheses
* p< 0.10, ** p< 0.05, *** p< 0.01

Back

# Theory

# Rate dispersion

**Limit cases of $\pi = \mathbf{0}, \mathbf{1}$** [3]

- Banks operate under monopoly/perfect competition

$\rightarrow$ degenerate distribution

Intermediate case $\pi \in (\mathbf{0}, \mathbf{1})$

- at any $r^*$, a bank deviating to $r^* - \varepsilon$ gains all shoppers $\rightarrow$ Strict gain
- at $r^* = c$, $\Pi^* = 0$. A bank deviating to any $r > c$ makes profits from captive firms

$\rightarrow$ dispersed equilibrium

**Isoprofit function**, top of the support at $\bar{r}^g$ (pooling, $\phi^{s_i} > 0$):

$$(1-\pi)(\bar{r}^g - c^{avg,s_i}) = [1 - \pi + \pi(1 - F^{s_i}(r))]\,(r - c^{avg,s_i})$$

[3] $\pi$ is the share of shoppers

# Where does the bank lend?

It depends on the severity of adverse selection $\phi^{s_i}$.

$$\boxed{\phi^{s_i} \equiv \left(\bar{r}^g - c^{avg,s_i}\right) - \mu_b\left(\bar{r}^b - c_b\right)}$$

Intuition: compares profits in pooling vs profits when lending to bad types only

$$\begin{cases} \phi^{s_i} > 0 & \text{mild: the bank } \textbf{pools} \text{ both types} \\ \underline{\phi}^{s_i} < \phi^{s_i} \leq 0 & \text{severe: a } \textbf{mix} \text{ of pooling and bad-only} \\ \phi^{s_i} \leq \underline{\phi}^{s_i} & \text{severe: } \textbf{bad types only} \end{cases}$$

with $\underline{\phi}^{s_i} = -\dfrac{\pi}{1-\pi}\left(\bar{r}^g - c^{avg,s_i}\right) < 0$, found from $F^a(\bar{r}^g) > 0$.

# Equilibrium under a perfect signal

**Proposition 1 (Dispersed-rate equilibrium, $q = 1$)**
*For $\pi \in (0,1)$ and a perfect signal there is a unique symmetric mixed-strategy equilibrium. For each type $i$, banks randomize over $[\underline{r}^i, \bar{r}^i]$ according to*

$$F^i(r) = 1 - \frac{1-\pi}{\pi}\left[\frac{\bar{r}^i - r}{r - c_i}\right],$$

*with $\bar{r}^i$ the type's reservation rate and $\underline{r}^i$ defined by $F^i(\underline{r}^i) = 0$.*

Observed rates are the mixture $\mu_g F^g(r) + \mu_b F^b(r)$ on $[\underline{r}^g, \bar{r}^b]$.

- the bank solves two separate problems, one by type, one per type $\rightarrow$ *between-type* dispersion

⏩ Derivation ⏩ Example

◀ Back

# Example: two types, perfect signal ($q = 1$)

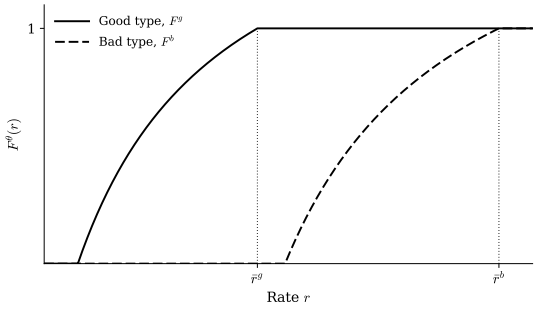



**Figure 5:** Equilibrium under a perfect signal.

- risky types cost more ($c_b$) and tolerate more ($\bar{r}^b$)
- With $q = 1$ the bank solves **two separate problems**, one per type
- observed rates are the mixture $\mu_g f^g(r) + \mu_b f^b(r)$

⏩ Formula ⏩ Proposition

# Equilibrium without information: two regimes

**Proposition 2 (Severe adverse selection, $\phi \leq 0$)**
*Banks randomize over $[\underline{r}^a, \bar{r}^b]$, mixing a pooling and a bad-only region: $F = F^a$ on $[\underline{r}^a, \bar{r}^g]$ and $F = F^a(\bar{r}^g) + F^b$ on $(\bar{r}^g, \bar{r}^b]$, with*

$$F^a(r) = \max\left\{1 - \frac{1-\pi}{\pi}\left[\frac{\mu_b(\bar{r}^b - c_b)}{r - c^{avg}(r)} - 1\right], 0\right\}$$

$$F^b(r) = 1 - F^a(\bar{r}^g) - \frac{1-\pi}{\pi}\left[\frac{\bar{r}^b - c_b}{r - c_b(r)} - 1\right]$$

**Proposition 3 (Mild adverse selection, $\phi > 0$)**
*Banks randomize over $[\underline{r}, \bar{r}^g]$ only, at $F(r) = 1 - \frac{1-\pi}{\pi}\left[(\bar{r}^g - r)/(r - c^{avg}(r))\right]$: all firms trade, both types face the same rates.*

**Lemma 1 (When does pooling survive?)**
*A pooling region exists for $-\frac{\pi}{1-\pi}(\bar{r}^g - c^{avg}) < \phi \leq 0$; below that threshold the bank lends to bad types only.*

Example

# Example: the three regimes within a signal group

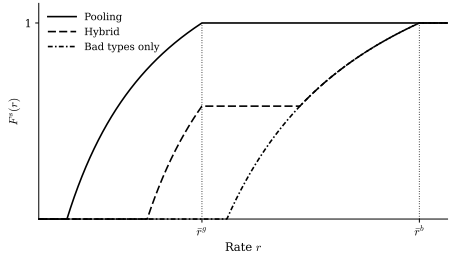



**Figure 6:** The three regimes within a signal group. Pooling, $\phi^s > 0$. Hybrid, $\underline{\phi}^s < \phi^s \leq 0$. Bad types only, $\phi^s \leq \underline{\phi}^s$.

- Possible outcomes: pooling, bad-only lending, or a mix, depending on the degree of adverse selection $\phi$
- With no signal ($q = 0.5$) both types face the **same** rates $\rightarrow$ no between-type dispersion

⏩ Other regimes ⏩ Proposition

# General case: informative signal, $q \in [0.5, 1)$

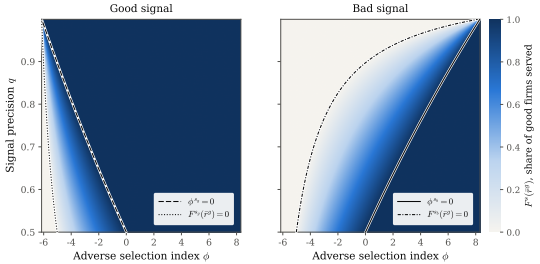
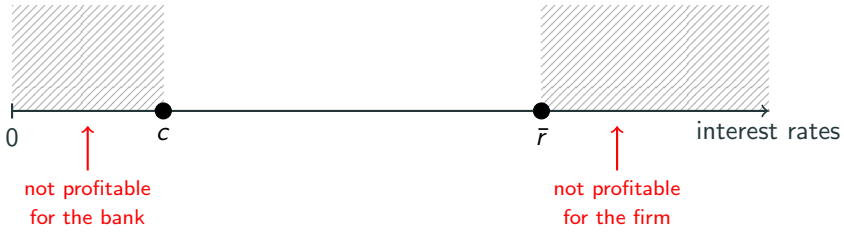



**Figure 7:** Share of good firms served, by signal group. Below $\phi^s = 0$: pooling. Below $F^s(\bar{r}^g) = 0$: no good type gets credit. Between the lines: hybrid.

- $q = 1/2$: the two groups merge, $\alpha^{s_g} = \alpha^{s_b} = \mu_g$
- $q \to 1$: $\phi^{s_g} \to \bar{r}^g - c_g > 0$, the good-signal group pools at the good types' cost; $\phi^{s_b} \to \bar{r}^g - \bar{r}^b < 0$, the bad-signal group lends to bad types only

# Search frictions only

**Example: feasible rates**




- Bank profits: $\Pi(r) = (1 - \pi + \pi(1 - F(r)))(r - c)$
- $\pi \to 0$ (monopoly) $r* = \bar{r}$ , $\pi \to 1$ $r* = c$ (Bertrand)
- for $\pi \in (0, 1)$: mixed strategy equilibrium

Formula Back

# $\phi$: how severe is adverse selection?

**Lemma 2 (Monopoly benchmark)**
*Under $\pi = 0$, $c'(r) < 1$ and an uninformative signal, define*

$$\boxed{\phi \;\equiv\; \big(\bar{r}^{g} - c^{avg}(\bar{r}^{g})\big) \;-\; \mu_b\big(\bar{r}^{b} - c_b(\bar{r}^{b})\big)}$$

*The unique equilibrium is $\bar{r}^{g}$ if $\phi > 0$, and $\bar{r}^{b}$ if $\phi \leq 0$.*

- $\phi > 0$: adverse selection is **mild** — the bank earns more by **pooling** both types
- $\phi \leq 0$: adverse selection is **severe** — the bank earns more from **bad types only**, and good types walk away

**Lemma 3 (Perfect competition)**
*Under $\pi = 1$ the unique equilibrium is a pooling contract at the competitive rate $r^{c} = c^{avg}$, accepted by the whole population; profits are zero.*

Back

# No Info - Monopoly

**If monopoly** $\pi = 0$

$$Profits = \begin{cases} (r - c^{avg}) & \text{if } \underline{r} < r \leq \bar{r}^g \\ \mu_b(r - c_b) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

The bank evaluates the maximum profit it can extract in each region. It compares:

$$\max\{(1-\pi)(\bar{r^g} - c_{avg}), \mu_b(1-\pi)(\bar{r^b} - c_b)\}$$

Define $\phi$:

$$\boxed{\phi = (\bar{r^g} - c_{avg}) - \mu_b(\bar{r^b} - c_b)}$$

1. $\phi \leq 0$ (strong AS) $\rightarrow$Banks extract more profits from bad types only
2. $\phi > 0$ (weak AS) $\rightarrow$Banks extract more profits from pooling good and bad types

# No Info - Perfect competition

**If perfect competition** $\pi = 1$

1. Banks make zero profits, offering the average cost in the population $c^{avg} = \mu_g c_g + \mu_b c_b$
2. As I consider the case $c^{avg} < \bar{r^g}$, the pooling rate will be always accepted by both types

$$Profits = (\mu_b + \mu_g)(c^{avg} - c^{avg}) = 0$$

# If p is a decreasing function of r

$$p = p(r) = \bar{p}^i - \gamma r$$

**Case 1: $\gamma$ is sufficiently small**

- Firm: cares about profit if success R-r, same $\bar{r}$
- Bank requires compensation for risk $\uparrow c = \frac{r^{CB}}{p(r)}$

**The equilibrium distribution F(r):**

$$F(r) = 1 - \frac{1-\pi}{\pi}\frac{\bar{r} - r}{r - \frac{r^{CB}}{p(r)}}, \qquad \text{if } \underline{r(\pi)} < r \leq \bar{r}$$

$$\underline{r(\pi)} = (1-\pi)(\bar{r} - \frac{r^{CB}}{p(r)}) + \frac{r^{CB}}{p(r)}$$

# If p is a decreasing function of r

**Case 2: $\gamma$ is sufficiently high**

- As before, banks require compensation for risk $\rightarrow\uparrow$c
- p(r) may go to zero if r is too high $\rightarrow \bar{r} < R$
- convex cost of lending $\rightarrow$Interior rate r* that maximizes profits

$\rightarrow$ The interval of feasible rates is smaller

**Equilibrium distribution $F(r)$:**

$$F(r) = 1 - \frac{1-\pi}{\pi} \frac{r^* - r}{r - \frac{r^{CB}}{p(r)}}, \quad \text{for } \underline{r}(\pi) < r \leq \bar{r}$$

# Perfect vs asymmetric information

1. **Perfect information / perfect signal q=1:**
   $\rightarrow$The bank can perfectly identify each firm's type
   $\rightarrow$Positive correlation between rates and riskiness

2. **Asymmetric information / q=0.5**
   $\rightarrow$The bank cannot distinguish types

3. **Imperfect public signal,** $q \in (0.5, 1)$, of a precision the market sets
   $\rightarrow$extreme precision $q_\theta$ values coincide with no/perfect information cases

# Perfect information - Density

Duplicate the one-firm problem: one distribution per type, and $F(r)$ is their sum.

$$f^i(r) = \begin{cases} \dfrac{1-\pi}{\pi} \dfrac{\bar{r}^i - c_i}{(r-c_i)^2}, & \text{if } \underline{r}^i(\pi) < r \leq \bar{r}^i, \\ 0, & \text{otherwise.} \end{cases}$$

Example

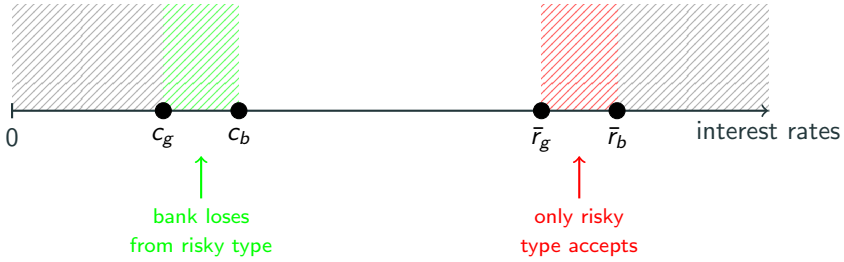
**Example: feasible rates**




- Hp: bad type has higher default probability but higher reservation rate (delivers consistent correlation patterns)
- the bank cannot perfectly recognize types, but knows $\mu_{bad}$, $\pi$, $\bar{r}_{bad}$, $\bar{r}_{good}$, $p_g$, $p_b$

# Asymmetric information

**Example: feasible rates**




- Hp: risky type has higher default probability but higher reservation rate (delivers consistent correlation patterns)
- the bank cannot perfectly recognize types, but knows $\mu_{bad}$, $\pi$, $\bar{r}_{bad}$, $\bar{r}_{good}$, $p_g$, $p_b$

# No info - $\phi \leq 0$

**Case 1. Banks extract more profits at $\bar{r}^b$.**

The **CDF** on both regions can be expressed as:

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}^a \\ F^a(r) & \text{if } \underline{r}^a \leq r \leq \bar{r}^g \\ F^a(\bar{r}^g) + F^b(r) & \text{if } \bar{r}^g < r \leq \bar{r}^b \\ 1 & \text{if } r > \bar{r}^b \end{cases}$$

Then, the **equal profit condition**:

$$[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}) = \mu_b(1 - \pi)(\bar{r}^b - c_b)$$
$$\mu_b[1 - \pi + \pi(1 - F^a(\bar{r^g}) - F^b(r))](r - c_b) = \mu_b(1 - \pi)(\bar{r}^b - c_b)$$

⏩ solution ⏩ back

# No info - $\phi \leq 0$: solution

**CDF**

$$F^a(r) = 1 - \frac{1-\pi}{\pi}\left[\frac{\mu_b(\bar{r}^b - c_b)}{(r - c^{avg})} - 1\right]$$

$$F^b(r) = 1 - F^a(\bar{r^g}) - \frac{1-\pi}{\pi}\left[\frac{(\bar{r}^b - c_b)}{(r - c_b)} - 1\right]$$

**Density**

$$f(r) = \begin{cases} \dfrac{1-\pi}{\pi}\,\dfrac{\mu_b(\bar{r}^b - c_b)}{(r - c^{avg})^2}, & \text{if } \underline{r}^a \leq r \leq \bar{r}^g, \\ \dfrac{1-\pi}{\pi}\,\dfrac{(\bar{r}^b - c_b)}{(r - c_b)^2}, & \text{if } \underline{r}^b < r \leq \bar{r}^b, \\ 0, & \text{otherwise.} \end{cases}$$

back

# No info - $\phi > 0$

**Case 2. Banks extract more profits at $\bar{r}^g$.**

**Intuition:** No contract on bad-types-only region is feasible. Full pooling equilibrium.

The **CDF** on both regions can be expressed as:

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}^a \\ F^a(r) & \text{if } \underline{r}^a \leq r \leq \bar{r}^g \\ 1 & \text{if } r > \bar{r}^g \end{cases}$$

Then, the **equal profit condition**:

$$[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}) = (1 - \pi)(\bar{r}^g - c^{avg})$$

so that

$$F^a(r) = 1 - \frac{1 - \pi}{\pi}\left[\frac{(\bar{r}^g - r)}{(r - c^{avg})}\right]$$

# The public signal and Bayesian updating

Both banks observe the same signal s and update the belief (unconditional probability) using Bayes' rule.

**Conditional probabilities:** By Bayes' rule, the probability that a firm is good given a good signal is

$$\begin{aligned}\Pr(\theta = G \mid s = G) &= \frac{Pr(\theta = G \cap s = G)}{Pr(s = G)} \\ &= \frac{\Pr(s = G \mid \theta = G)\Pr(\theta = G)}{\Pr(s = G \mid \theta = G)\Pr(\theta = G) + \Pr(s = G \mid \theta = B)\Pr(\theta = B)} \\ &= \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b)\mu_b}\end{aligned}$$

# Signal Precision & Bayesian Updating

**Signal precision:** For each type $\theta \in \{g, b\}$, both banks observe the same binary signal $s$ with

$$\Pr(s = \theta \mid \theta) = q_\theta \geq 0.5$$

$q_\theta = \frac{1}{2}$: uninformative (no information); $q_\theta = 1$: perfect information.

**Unconditional signal probabilities:**

$$\sigma_g \equiv \Pr(s = g) = q_g \mu_g + (1 - q_b)\mu_b, \quad \sigma_b \equiv \Pr(s = b) = q_b \mu_b + (1 - q_g)\mu_g$$

**Bayesian updating:**

$$\alpha^{s_g} \equiv \Pr(\theta = g \mid s = g) = \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b)\mu_b},$$

$$\alpha^{s_b} \equiv \Pr(\theta = g \mid s = b) = \frac{(1 - q_g)\mu_g}{(1 - q_g)\mu_g + q_b \mu_b}$$

$$\alpha^{s_g} > \mu_g > \alpha^{s_b}$$

# Pricing on the signal

**Expected outcome:**

1. Following the same procedure as before, banks compare profits at the two reservation prices using the updated probabilities
2. now, conditional on a good signal, the average population of good types met by the bank is higher than $\mu_g$, and vice-versa for bad signals
3. in equilibrium, two distributions, one for each signal realization

Outcome:

- $\uparrow$informative signal $\rightarrow\downarrow$adverse selection problem, because the average cost and population share are closer to the true ones
- $\uparrow$informative signal $\rightarrow\uparrow$dispersion
- $\uparrow$informative signal $\rightarrow\uparrow$correlation between ex-ante rate and ex-post default probability

# One type: equilibrium distribution

**Equal-profit condition** (highest rate must be the reservation rate $\bar{r}$):

$$(1-\pi)(\bar{r}-c) \;=\; \big[1-\pi+\pi(1-F(r))\big](r-c)$$

**CDF and density:**

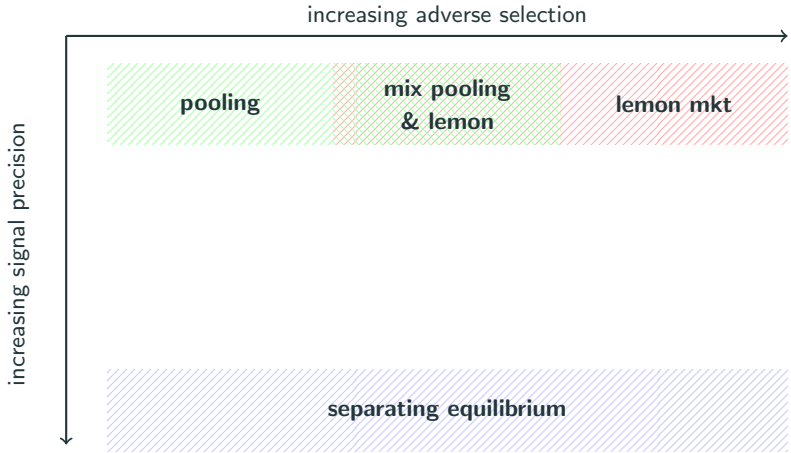
$$F(r) = 1 - \frac{1-\pi}{\pi}\left[\frac{\bar{r}-r}{r-c}\right], \qquad\qquad f(r) = \frac{1-\pi}{\pi}\,\frac{\bar{r}-c}{(r-c)^2}$$

on the support $\underline{r} \le r \le \bar{r}$, with $\underline{r} = (1-\pi)\bar{r} + \pi c$ from $F(\underline{r}) = 0$.

$f'(r) < 0$: the density is right-skewed — most quotes cluster near the competitive rate, but captive firms sustain a tail up to $\bar{r}$.

⏩ Set-up ⏩ Back

# Equilibrium: Overview




Back

# Calibration

# Calibration (moment matching)

**Objective:** recover parameters of competition $\pi$ and signal precision $q$

**Moments matched :** Mean: $\mathbb{E}[r]$, Variance: $\text{Var}(r)$, Correlation with default probability: $\text{Corr}(r, p)$

⏩ Details ⏩ Back

# Calibration: Parameters & Moments

**Parameters recovered (3):**

- $\pi$: search friction (share of shoppers visiting both banks)
- $q$: precision of the public signal available in the market
- $\bar{r}_g$: reservation rate for good types

**Moments matched (3):** mean $\mathbb{E}[r]$, variance $\text{Var}(r)$, correlation with default probability $\text{Corr}(r, p)$

**Calibrated from data (by rating class):**

- Success probabilities $p_g, p_b$ and type shares $\mu_b$ from rating transitions
- Upper bound bad types $\bar{r}_b$ from observed $\max(r)$
- Fixed: $r_{CB} = 1$ (average policy rate over the sample period)

data moments Back

# Calibration: Strategy

**Objective:**

minimize squared proportional errors across three moments:

$$Q(\theta) = \sum_{m} \left( \frac{m^{\text{model}} - m^{\text{data}}}{m^{\text{data}}} \right)^2$$

**Numerical implementation:**

Grid search ($20^3$ combinations over $\pi, q, \bar{r}_g$) + L-BFGS-B local refinement

# Results: public information across rating classes

**Table 1:** Search Frictions and Information by Rating Class

| Rating | $\hat{\pi}$ | $\hat{q}$ | $\hat{\bar{r}}_g$ | Info Contr. (%) | Obj. Value $Q(\theta)$ |
|---|---|---|---|---|---|
| 3+ | 0.772 | 0.997 | 3.461 | 4.89 | 0.41 |
| 3 | 0.759 | 1.000 | 3.706 | 4.56 | 0.35 |
| 4+ | 0.738 | 0.999 | 4.609 | 3.94 | 0.15 |
| 4 | 0.757 | 1.000 | 5.056 | 5.06 | 0.19 |
| 5+ | 0.749 | 1.000 | 5.403 | 3.05 | 0.20 |
| 5 | 0.729 | 0.944 | 6.662 | 0.04 | 0.04 |
| 6 | 0.721 | 0.526 | 7.057 | 0.00 | 0.04 |
| 7 | 0.711 | 0.526 | 9.350 | 0.00 | 0.00 |

Notes: $\hat{\pi}$ is the search-friction parameter, $\hat{q}$ the signal precision, $\hat{\bar{r}}_g$ the reservation rate of good borrowers, and "Information Contribution" measures the percentage increase in rate dispersion attributable to screening relative to the no-information benchmark.

**Key findings:**

1. **Public information differs across classes:** the market's signal is close to perfect in the safe classes (3+ to 5+, $\hat{q} \geq 0.99$), less precise in class 5 ($\hat{q} = 0.94$), and uninformative in the risky ones (6 and 7, $\hat{q} \approx 0.53$).
2. **Competition is similar across classes:** $\hat{\pi} \in [0.71, 0.77]$.

⏩ Back ⏩ Fit ⏩ Bank surplus ⏩ Surplus sharing

# The model fits the correlation; variance only in part

**Table 2:** Model Fit: Observed vs. Model-Implied Moments

| | Mean | | Variance | | Correlation | |
|---|---|---|---|---|---|---|
| Rating | Data | Model | Data | Model | Data | Model |
| 3+ | 1.67 | 1.94 | 0.406 | 0.154 | 0.025 | 0.025 |
| 3 | 1.83 | 2.08 | 0.443 | 0.188 | 0.033 | 0.033 |
| 4+ | 2.40 | 2.53 | 0.553 | 0.338 | 0.032 | 0.032 |
| 4 | 2.37 | 2.63 | 0.730 | 0.424 | 0.048 | 0.048 |
| 5+ | 2.57 | 2.80 | 0.867 | 0.492 | 0.048 | 0.048 |
| 5 | 3.36 | 3.44 | 0.985 | 0.794 | 0.017 | 0.017 |
| 6 | 3.63 | 3.67 | 1.137 | 0.913 | 0.000 | 0.000 |
| 7 | 4.79 | 4.78 | 1.843 | 1.725 | 0.000 | 0.000 |

Notes: Mean is the class mean rate and variance the variance of rate residuals within the class, both in percentage points. Correlation is between the rate residual and the three-year-ahead change in the borrower's default probability. For ratings 6 and 7 the estimated correlation is not significantly different from zero ($p = 0.65$ and $p = 0.54$) and is set to zero as a calibration target.

- At 3+, $c_b - c_g \approx 0.003$ against a residual s.d. of 0.64: search must supply the variance

Back

# Back to the data

**How to rationalize the differences in different markets?** ⏩ high vs low

**Low PD firms:** low dispersion, high correlation.

- Narrow rate region (parameters)
- High information
- Low/high competition?

**High PD firms:** high dispersion, no correlation

- Wider rate region (parameters, ex. $R^b \gg R^g$, $p^b \ll p^g$)
- Low information available in the segment $\rightarrow$pooling is more likely
- Low/high competition?

$\rightarrow$ Calibration results are consistent with these patterns (see Calibration section).

# Correlation by type

Imagine that within each rating class there are good and bad firms, that you can recognize by looking at the evolution of their default probability.

1. good if PD increases after 10 years, bad if PD decreases, null otherwise
2. Collect residuals and study them separately in each sample

Good rating market:

- good (7th) and bad (28) types .
  Mean: $mean^{good} = -0.04 < mean^{bad} = 0.02$. SD= $sd^{good} = 0.65 < sd^{bad} = 0.78$, correlation is positive

Bad rating market:

- good (6th) and bad (13) types .
  Mean: $mean^{good} = -0.057 < mean^{bad} = 0.05$. SD= $sd^{good} = 0.98 < sd^{bad} = 1.19$, correlation is positive

# Identification: full rank, weakest along $q$

**Table 3:** Local Identification Diagnostics: Elasticity-Scaled Jacobian of (mean, var, corr) w.r.t. $(\pi, q, \bar{r}_g)$

| Rating | Rank | Condition Number | Weakest Direction |
|---|---|---|---|
| 3+ | 3/3 | 431.7 | $q$ |
| 3 | 3/3 | 209.4 | $q$ |
| 4+ | 3/3 | 134.2 | $q$ |
| 4 | 3/3 | 82.1 | $q$ |
| 5+ | 3/3 | 65.9 | $q$ (mixed with $\bar{r}_g$) |
| 5 | 3/3 | 4.8 | $\pi$ (mixed with $\bar{r}_g$) |
| 6 | 3/3 | 21.6 | $q$ |
| 7 | 3/3 | 14.1 | $q$ |

Notes: Rank and condition number computed from the singular value decomposition of the elasticity-scaled Jacobian (percentage change in moment per percentage change in parameter) at each rating class's estimated $\hat{\theta}$. "Weakest direction" identifies the parameter with the largest loading on the smallest singular vector; a second parameter is named when its loading also exceeds 0.45 in absolute value.

- A dense global search and a regime-by-regime fit return the baseline estimate

Back

# Calibration with $\pi$ fixed at multi-bank relationships

**Table D.11:** Alternative Specification: $\pi$ Fixed at the Share of Firms with At Least Two Bank Relationships

| Rating | $\pi$ (fixed) | $\hat{q}$ | $\hat{\bar{r}}_g$ | Obj. (fixed $\pi$) | Obj. (baseline) | Var. reachable? |
|---|---|---|---|---|---|---|
| 3+ | 0.600 | 0.500 | 3.055 | 0.62 | 0.41 | No |
| 3 | 0.651 | 0.985 | 3.709 | 0.45 | 0.35 | Yes |
| 4+ | 0.651 | 0.986 | 4.609 | 0.21 | 0.15 | Yes |
| 4 | 0.675 | 1.000 | 5.080 | 0.25 | 0.19 | Yes |
| 5+ | 0.670 | 0.999 | 5.417 | 0.25 | 0.20 | Yes |
| 5 | 0.562 | 0.991 | 6.662 | 0.22 | 0.04 | Yes |
| 6 | 0.399 | 0.552 | 6.347 | 0.64 | 0.04 | Yes |
| 7 | 0.426 | 0.554 | 8.895 | 0.47 | 0.00 | Yes |

Notes: $\pi$ is fixed at the share of firms in each rating class with at least two active bank relationships, averaged over the sample period; $q$ and $\bar{r}_g$ are then re-estimated with $r_{CB}$ fixed at 1, as in the baseline. "Obj. (baseline)" reproduces the objective value from Table 1; both runs match the same moments, so the two columns are directly comparable. "Var. reachable?" indicates whether the observed variance falls inside the model's achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed, lower $\pi$.

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# Defaults as bad types: inputs

**Table E.12:** Inputs that change under the alternative bad-type definition

| | $\mu_b$ | | $p_b$ | | Rate–default corr. | |
|---|---|---|---|---|---|---|
| Rating | Baseline | With default | Baseline | With default | Baseline | With default |
| 3+ | 0.846 | 0.846 | 0.997 | 0.997 | 0.025 | 0.023 |
| 3 | 0.749 | 0.751 | 0.996 | 0.996 | 0.033 | 0.034 |
| 4+ | 0.739 | 0.745 | 0.992 | 0.992 | 0.032 | 0.022 |
| 4 | 0.671 | 0.691 | 0.981 | 0.981 | 0.048 | 0.037 |
| 5+ | 0.491 | 0.549 | 0.965 | 0.965 | 0.048 | 0.044 |
| 5 | 0.268 | 0.415 | 0.908 | 0.630 | 0.017 | 0.049 |
| 6 | 0.066 | 0.344 | 0.784 | 0.200 | 0.000 | 0.068 |
| 7 | 0.069 | 0.446 | 0.630 | 0.200 | 0.000 | 0.000 |

Notes: "Baseline" is the definition used in the calibration section, in which bad types are firms whose rating deteriorates; "With default" adds firms that default. For ratings 6 and 7 the raw data give a default frequency of exactly one in the bad group, which would imply an infinite cost of lending $c_b = r_{CB}/p_b$; $p_b$ is therefore floored at 0.2, so the reported values for those two classes are the floor rather than an estimate.

# Defaults as bad types: estimates

**Table E.13:** Search frictions and information by rating class, defaults counted as bad

| Rating | $\hat{\pi}$ | $\hat{q}$ | $\hat{\bar{r}}_g$ | Info Contr. (%) | Obj. Value $Q(\theta)$ |
|---|---|---|---|---|---|
| 3+ | 0.772 | 0.988 | 3.461 | 4.58 | 0.41 |
| 3 | 0.759 | 1.000 | 3.703 | 4.76 | 0.35 |
| 4+ | 0.737 | 0.978 | 4.609 | 2.83 | 0.16 |
| 4 | 0.758 | 1.000 | 5.092 | 3.81 | 0.19 |
| 5+ | 0.750 | 1.000 | 5.416 | 2.99 | 0.19 |
| 5 | 0.737 | 0.757 | 6.662 | 1.01 | 0.06 |
| 6 | 0.767 | 0.613 | 7.057 | 12.79 | 0.24 |
| 7 | 0.754 | 0.526 | 9.350 | 0.37 | 0.20 |

Notes: Columns as in Table 1, which reports the same quantities under the baseline definition.

◂ Back

# Defaults as bad types: fit

**Table E.14:** Model fit, defaults counted as bad

| | Mean | | Variance | | Correlation | |
|---|---|---|---|---|---|---|
| Rating | Data | Model | Data | Model | Data | Model |
| 3+ | 1.67 | 1.94 | 0.406 | 0.153 | 0.023 | 0.023 |
| 3 | 1.83 | 2.08 | 0.443 | 0.188 | 0.034 | 0.034 |
| 4+ | 2.40 | 2.53 | 0.553 | 0.334 | 0.022 | 0.022 |
| 4 | 2.37 | 2.63 | 0.730 | 0.426 | 0.037 | 0.037 |
| 5+ | 2.57 | 2.81 | 0.867 | 0.494 | 0.044 | 0.043 |
| 5 | 3.36 | 3.51 | 0.985 | 0.740 | 0.049 | 0.049 |
| 6 | 3.63 | 4.15 | 1.137 | 0.600 | 0.068 | 0.070 |
| 7 | 4.79 | 5.39 | 1.843 | 1.064 | 0.000 | 0.003 |

Notes: Moments defined as in Table 2.

# Surplus sharing

# Who gets the surplus?

**Proposition 4**
*The banks' share of total surplus is decreasing in competition and increasing in the precision of the public signal.*

1. **Under a perfect signal**, every firm trades, each type is priced off its own cost, and the bank captures exactly $1-\pi$ of the realized surplus, firms $\pi$.
2. **Under no information**, the bank collects strictly less:

$$\mathbb{E}[\Pi(q)] \;\propto\; (1-\pi), \qquad \Pi^{LOSS} \;=\; \begin{cases} (1-\pi)\,\mu_b\,(\bar{r}^b - \bar{r}^g) & \text{if } \phi > 0 \\ (1-\pi)\,\mu_g\,(\bar{r}^g - c_g) & \text{if } \phi \leq 0 \end{cases}$$

Banks recover the loss from adverse selection at q=1 → information is worth more to banks where competition is low

⏩ Gains by class

# Who gets the surplus?

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Banks recover the loss from adverse selection at q=1 → information is worth more to banks where competition is low

⏩ Gains by class

# . . . and what do firms get?

**Proposition 5 (Firms' surplus and information)**
*Information reduces the costs borne by good types.*

**Under no information,** *good types*

- *when adverse selection is mild ($\phi > 0$), trade at the higher* **pooling** *rates* $\rightarrow$ **cross-subsidy** *to bad types*
- *when it is severe ($\phi \leq 0$)* **may not trade at all** $\rightarrow$ **exit** *from the market*

*Their gains rise with $q$, fall when adverse selection is severe and fall with competition.*

Good types bear two costs of asymmetric information — a **cross-subsidy** to bad types where the bank pools, and **exclusion** where it lends to bad types only. Both fall with $q$.

⏩ Gains by class ⏩ Back

# . . . and what do firms get?

**Proposition 6 (Firms' surplus and information)**
*Information increases the costs borne by bad types.*

**Under no information,** *bad types*

- *when* $\phi > 0$, **benefit** *from pooling* $\rightarrow$ *benefit from* **cross-subsidy** *from good types and from banks*
- *when it is severe* ($\underline{\phi} < \phi \leq 0$)*, they may gain if* **partial pooling** $\rightarrow$ **cross-subsidy** *from good types*
- *when it is severe* ($\phi \leq \underline{\phi}$)*, same as perfect information* $\rightarrow$ **no gain**

*Their benefits fall with q, fall when adverse selection is severe and fall with competition..*

⏩ Gains by class ⏩ Back

# Redistribution: percent of total surplus

**Table E.15:** Redistribution from a perfect signal, defaults counted as bad: percent of the rating class's total surplus

| Rating | Bad types | Banks | Good types |
|---|---|---|---|
| 3+ | -0.893 | 0.883 | 0.010 |
| 3 | -0.881 | 0.868 | 0.012 |
| 4+ | -0.817 | 0.795 | 0.020 |
| 4 | -0.747 | 0.693 | 0.052 |
| 5+ | -0.640 | 0.547 | 0.094 |
| 5 | -1.901 | 0.430 | 1.471 |
| 6 | -12.231 | 0.396 | 11.834 |
| 7 | -9.450 | 0.506 | 8.942 |

Notes: change in each party's expected surplus when signal precision moves from $q = 0.5$ to $q = 1$, as a percentage of the class's maximum total surplus. Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information redistributes without creating surplus.

Back

% of own surplus

# Redistribution: percent of each party's surplus

**Table E.16:** Redistribution from a perfect signal, defaults counted as bad: percent of each party's own surplus

| Rating | Bad types | Banks | Good types |
|---|---|---|---|
| 3+ | -1.65 | 2.43 | 0.10 |
| 3 | -1.87 | 2.29 | 0.08 |
| 4+ | -1.83 | 1.94 | 0.14 |
| 4 | -1.73 | 1.81 | 0.28 |
| 5+ | -1.89 | 1.39 | 0.35 |
| 5 | -7.61 | 1.04 | 4.35 |
| 6 | -55.09 | 1.06 | 29.28 |
| 7 | -33.82 | 1.30 | 27.02 |

Notes: the same changes, each divided by that party's own surplus at $q = 0.5$ rather than by the class total. Rows do not sum. This is the denominator relevant to a bank weighing the cost of acquiring precision, and to a borrower judging the size of the repricing.

# Stability of the calibrated precision

# From a public signal to information acquisition

**So far:** $q$ is a feature of the market, and every bank reads the same signal.

**Now:** each bank chooses whether to pay $C_s$ for the signal ($q = 1$) or to lend on the prior ($q = 0.5$)

- banks are atomistic: one bank's choice leaves the market's rate distribution unchanged

**Question:** is the calibrated pattern an equilibrium of this game?

- $\hat{q} \approx 1$ in classes 3+ to 5: no bank gains by **dropping** the signal, Condition B: $C_s \leq L$
- $\hat{q} \approx 0.5$ in classes 6 and 7: no bank gains by **acquiring** it, Condition A: $G \leq C_s$

# If information only sorts types, the informed equilibrium is unstable

**Deviation from informed equilibrium $\mathbf{q = 1}$**: a bank that stops acquiring the signal and posts $r = \bar{r}^g$ earns exactly the informed profit, for any rate on the common support of the perfect info distributions

$$\mu_g \, A_g(\bar{r}^g)\,(\bar{r}^g - c_g) \;+\; \mu_b \, A_b(\bar{r}^g)\,(\bar{r}^g - c_b) \;=\; \mu_g \bar{\Pi}_g + \mu_b \bar{\Pi}_b,$$

so it loses nothing $L = 0$ and saves the cost $C_s > 0$. Needs $\bar{r}^g$ inside both types' supports, $\pi \geq (\bar{r}^b - \bar{r}^g)/(\bar{r}^b - c_b)$: true in every calibrated rating class.

**Deviation from uninformed equilibrium $\mathbf{q = 0.5}$**: the bank gains

$$G = \pi \mu_g \mu_b (c_b - c_g) + (1 - \pi)\mu_b (\bar{r}^b - \bar{r}^g) > 0$$

$\rightarrow$ only the uninformed equilibrium can survive, yet $\hat{q} \approx 1$ in the safe classes

Information must create value beyond recognizing types

# Information also lowers risk: monitoring

A bank that acquires information also **monitors** its borrowers

- repayment rises with $q$, so the cost of lending falls: $c_i(q) = \dfrac{r^{CB}}{p_i(q)}, \;\; p_i'(q) > 0$
- monitoring strength $\delta_j$ in rating class $j$: $c_i(1) = (1 - \delta_j)\, c_i(0.5)$
- monitoring saving: $\kappa_i \equiv c_i(0.5) - c_i(1) = \delta_j\, c_i(0.5) \geq 0$
- informed banks lend at a lower cost $\rightarrow$ **larger gains from trade** $\bar{r}^i - c_i(1)$

Gain from moving the whole market from $q = 0.5$ (pooling) to $q = 1$:

$$\frac{\Pi^{q=1} - \Pi^{q=0.5}}{1 - \pi} = \underbrace{\mu_b\left(\bar{r}^b - \bar{r}^g\right)}_{\text{sorting}} + \underbrace{\mu_g \kappa_g + \mu_b \kappa_b}_{\text{monitoring}}$$

With $\delta_j = 0$ only the sorting term is left, and the informed equilibrium unravels.

# With monitoring, the informed equilibrium can be stable

**Proposition 7 (Stability of the informed equilibrium)**
*A bank that drops the signal in a market at $q = 1$ pays the monitoring saving on every loan it still makes:*

$$L \;=\; \mu_g\,(1-\pi)\,\kappa_g \;+\; \mu_b\,\frac{\bar{\Pi}_b}{\bar{r}^g - c_b}\,\kappa_b.$$

*$q = 1$ is an equilibrium if and only if $C_s \le L$: the monitoring effect must be large enough.*

In a market at $q = 0.5$, a bank that acquires the signal gains

$$G = \pi\mu_g\mu_b(c_b - c_g) + (1-\pi)\mu_b(\bar{r}^b - \bar{r}^g) + \big[(1-\pi) + \pi\mu_g\big](\mu_g\kappa_g + \mu_b\kappa_b)$$

| | |
|---|---|
| $C_s < L$ | only the **informed** equilibrium survives |
| $L < C_s < G$ | neither survives: the equilibrium is interior |
| $C_s > G$ | only the **uninformed** equilibrium survives |

Costs at $q = 1$: $c_i \equiv c_i(1)$. $L$ assumes the deviator keeps lending to both types; $G$ assumes it prices bad types at $\bar{r}^b$, as in classes 6 and 7.

Coefficients

# One signal cost needs a ninefold gap in monitoring

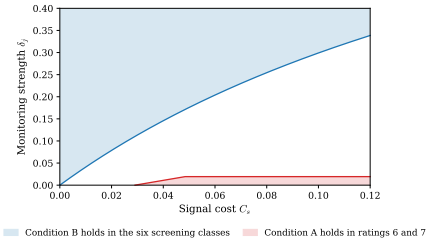



**Figure 9:** Monitoring strength $\delta_j$ that sustains the calibrated precision, against $C_s = C_s(1) - C_s(0.5)$, with $r^{CB} = 1$. Upper band: full information stable in the safest classes. Lower band: no information stable in the worse classes.

- The two bands never meet
- At $C_s = 0.048$ six classes need $\delta \geq 0.17$: a bad type's repayment probability rises 15 to 17 points
- Ratings 6 and 7 admit at most $\delta = 0.019$: 1 to 2 points

Ceiling Intervals

# Coefficients of the two conditions

**Table F.17:** Coefficients of the two stability conditions, by rating class

| Rating | Role | $c_b$ | $\bar{\Pi}_b/w$ | $K_j$ | $M_j$ |
|---|---|---|---|---|---|
| 3+ | screening | 1.003 | 0.235 | 0.234 | 0.087 |
| 3 | screening | 1.004 | 0.248 | 0.247 | 0.164 |
| 4+ | screening | 1.008 | 0.269 | 0.269 | 0.246 |
| 4 | screening | 1.020 | 0.252 | 0.252 | 0.324 |
| 5+ | screening | 1.036 | 0.258 | 0.260 | 0.562 |
| 5 | screening | 1.102 | 0.278 | 0.284 | 1.120 |
| | | | | $G_{0,j}$ | $S_j$ |
| 6 | not screening | 1.275 | - | 0.014 | 0.976 |
| 7 | not screening | 1.587 | - | 0.029 | 0.996 |

Notes: at the estimates of Table 1 and the calibrated $\mu_b$, $p_g$, $p_b$ and $\bar{r}^b$, with $r^{CB} = 1$. "Screening" classes are estimated at $\hat{q} \approx 1$ and Condition B applies; the other two at $\hat{q} = 0.53$ and Condition A applies. $c_b$ is the monitored cost $c_b(1)$ for a screening class and the un-monitored $c_b(0.5)$ for the other two. $K_j$ and $M_j$ are the coefficients of $L_j$ in $t = \delta_j/(1-\delta_j)$; $G_{0,j}$ and $S_j$ the intercept and the slope of $G_j$ in $\delta_j$. $K_j'$, the slope of the bad-only branch, runs from 0.080 at rating 5 to 0.195 at rating 4+ and never binds in the relevant range.

# Monitoring strength consistent with the estimates

**Table F.18:** Monitoring strength consistent with the estimated precision, at $C_s = 0.0483$

| Rating | Role | From | To | Upper end set by |
|---|---|---|---|---|
| 3+ | screening | **0.171** | 0.710 | pooling regime, $c_b(0.5) < \bar{r}^g$ |
| 3 | screening | 0.164 | 0.729 | pooling regime |
| 4+ | screening | 0.152 | 0.781 | pooling regime |
| 4 | screening | 0.161 | 0.798 | pooling regime |
| 5+ | screening | 0.157 | 0.808 | pooling regime |
| 5 | screening | 0.145 | 0.835 | pooling regime |
| 6 | not screening | 0 | **0.0193** | $p_g(1) \leq 1$; Condition A alone allows 0.0351 |
| 7 | not screening | 0 | **0.0193** | Condition A; $p_g(1) \leq 1$ allows 0.0347 |

Notes: lower ends from $\delta_j \geq C_s/(K_j + C_s)$, upper ends from $\delta_j \leq (C_s - G_{0,j})/S_j$ and from feasibility, at the estimates of Table 1. The binding pair is $\delta_j = 0.171$ in the six screening classes against $\delta_j = 0.0193$ in ratings 6 and 7, a ratio of 8.8. At those values the saving per unit lent is $\kappa_g \in [0.206, 0.210]$ and $\kappa_b \in [0.207, 0.227]$ in the screening classes, raising the bad type's repayment probability by 15.5 to 17.0 points, against $\kappa_g = 0.020$, $\kappa_b \in [0.025, 0.031]$ and a gain of 1.2 to 1.5 points in ratings 6 and 7.

# The ceiling in the two non-screening classes

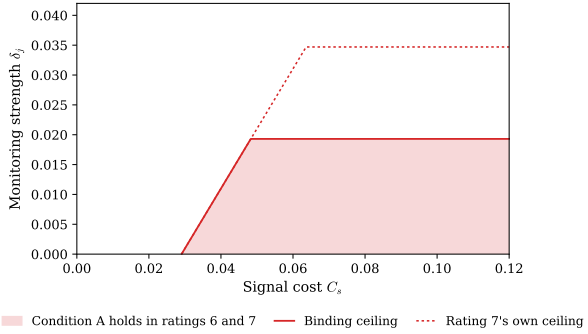



**Figure F.22:** Ceiling on monitoring strength in ratings 6 and 7. Solid: binding ceiling, the lower of class 7's Condition A and class 6's cap $p_g(1) \leq 1$. Dotted: class 7's ceiling, which continues past $C_s = 0.0483$ and reaches the cap 0.0347 at $C_s = 0.0636$.